

FIBERED CATEGORIES AND DESCENT

SGA 1, EXPOSÉ VI

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ABSTRACT. This is an English translation of Exposé VI, “Catégories fibrées et descente”, of SGA 1 (A. Grothendieck, *Revêtements étales et groupe fondamental*, 1960–61). The mathematical content follows the slightly corrected SMF recomposition (arXiv:math/0206203v2). Numbering of statements is that of the original exposé.

CONTENTS

0. Introduction	2
1. Universes, categories, equivalence of categories	2
2. Categories over another	3
3. Change of base in categories over $\mathcal{E}$	5
4. Fiber-categories; equivalence of $\mathcal{E}$ -categories	8
5. Cartesian morphisms, inverse images, cartesian functors	9
6. Fibered categories and prefibered categories. Products and change of base therein	11
7. Cloven categories over $\mathcal{E}$	13
8. Cloven category defined by a pseudofunctor $\mathcal{E}^\circ \rightarrow \mathbf{Cat}$	16
9. Example: cloven category defined by a functor $\mathcal{E}^\circ \rightarrow \mathbf{Cat}$; split categories over $\mathcal{E}$	18
10. Cofibered categories, bifibered categories	19
11. Various examples	19
12. Functors on a cloven category	23
References	25

English translation of Exposé VI of *Revêtements étales et groupe fondamental* (SGA 1), Séminaire de géométrie algébrique du Bois Marie, 1960–61, recomposed SMF edition, arXiv:math/0206203v2. The other exposés are translated in the sibling directories.

0. INTRODUCTION

Contrary to what had been announced in the introduction to the preceding exposé, it has turned out to be impossible to do descent in the category of preschemes, even in special cases, without having first developed with sufficient care the language of descent in general categories.

The notion of “descent” provides the general framework for all procedures of “gluing” of objects, and consequently of “gluing” of categories. The most classical case of gluing is relative to the datum of a topological space X and a covering of X by open sets X_i ; if one is given for every i a fiber space (say) E_i over X_i , and for every pair (i, j) an isomorphism f_{ji} of $E_i|X_{ij}$ onto $E_j|X_{ij}$ (where one sets $X_{ij} = X_i \cap X_j$), satisfying a well-known transitivity condition (which one writes in abbreviated form $f_{kj}f_{ji} = f_{ki}$), one knows that there exists a fiber space E over X , defined up to isomorphism by the condition that one have isomorphisms $f_i: E|X_i \xrightarrow{\sim} E_i$, satisfying the relations $f_{ji} = f_j f_i^{-1}$ (with the usual abuse of writing). Let X' be the sum space of the X_i , which is thus a fiber space over X (i.e. equipped with a continuous map $X' \rightarrow X$). One can interpret more concisely the datum of the E_i as a fiber space E' over X' , and the datum of the f_{ji} as an isomorphism between the two inverse images (by the two canonical projections) E'_1 and E'_2 of E' on $X'' = X' \times_X X'$, the gluing condition then being able to be written as an identity between isomorphisms of fiber spaces E'_1 and E'_3 over the triple fibered product $X''' = X' \times_X X' \times_X X'$ (where E'_i denotes the inverse image of E' on X''' by the canonical projection of index i). The construction of E , starting from E' and from f , is a typical case of a procedure of “descent”. Moreover, starting from a fiber space E over X , one says that X is “locally trivial”, of fiber F , if there exists an open covering (X_i) of X such that the $E|X_i$ are isomorphic to $F \times X_i$, or what amounts to the same, such that the inverse image E' of E on $X' = \coprod_i X_i$ is isomorphic to $X' \times F$.

Thus, the notion of “gluing” of objects like that of “localization” of a property, are tied to the study of certain types of “changes of base” $X' \rightarrow X$. In Algebraic Geometry, many other types of change of base, and notably the faithfully flat morphisms $X' \rightarrow X$, must be regarded as corresponding to a procedure of “localization” relative to preschemes, or other objects, “over” X . This type of localization is used just as much as simple topological localization (which is moreover a special case of it). The same holds (to a lesser extent) in Analytic Geometry.

Most of the proofs, reducing to verifications, are omitted or merely sketched; if need be we specify the less evident diagrams which arise in a proof.

1. UNIVERSES, CATEGORIES, EQUIVALENCE OF CATEGORIES

To avoid certain logical difficulties, we shall admit here the notion of *Universe*, which is a set “large enough” that one does not leave it by the usual operations of set theory; an “*axiom of Universes*” guarantees that every object lies in a Universe. For details, see a book in preparation by C. Chevalley and the speaker¹. Thus, the symbol **Ens** designates, not the category of all sets (a notion which has no meaning), but the category of sets which lie in a given Universe (which we shall not specify here in the notation). Likewise, **Cat** will designate the category of categories lying in the Universe in question, the “morphisms” from an object X of **Cat** to another Y being by definition the *functors* from X to Y .

If $\mathcal{C}$ is a category, we denote by $\text{Ob}(\mathcal{C})$ the set of objects of $\mathcal{C}$, by $\text{Fl}(\mathcal{C})$ the set of arrows of $\mathcal{C}$ (or morphisms of $\mathcal{C}$). We shall thus write $X \in \text{Ob}(\mathcal{C})$, avoiding the common abuse of notation $X \in \mathcal{C}$. If $\mathcal{C}$ and $\mathcal{C}'$ are two categories, a *functor* from $\mathcal{C}$ to $\mathcal{C}'$ will always be what is commonly called a *covariant* functor from $\mathcal{C}$ to $\mathcal{C}'$; its datum implies that of the target category and the source category, $\mathcal{C}$ and $\mathcal{C}'$. The functors from $\mathcal{C}$ to $\mathcal{C}'$ form a set, denoted

¹The definitive authors are C. Chevalley and P. Gabriel. The book is to appear in the year 3000. In the meantime, cf. also SGA 4 I.

$\text{Hom}(\mathcal{C}, \mathcal{C}')$, which is the set of objects of a category denoted $\mathbf{Hom}(\mathcal{C}, \mathcal{C}')$. By definition, a *contravariant functor* from $\mathcal{C}$ to $\mathcal{C}'$ is a functor from the *opposite category* $\mathcal{C}^\circ$ of $\mathcal{C}$ to $\mathcal{C}'$.

We shall admit the notion of *projective limit* and of *inductive limit* of a functor $F: \mathcal{I} \rightarrow \mathcal{C}$, and in particular the most common special cases of this notion: cartesian products and fibered products, dual notions of direct sums and of amalgamated sums, and the usual formal properties of these operations.

For example, in the category $\mathbf{Cat}$ introduced above, the projective limits (relative to categories $\mathcal{I}$ lying in the chosen Universe) exist; the set of objects (resp. the set of arrows) of the projective limit category $\mathcal{C}$ of the $\mathcal{C}_i$, is obtained by taking the projective limit of the sets of objects (resp. of the sets of arrows) of the categories $\mathcal{C}_i$. The best-known case is that of the product of a family of categories. We shall constantly use in the sequel the fibered product of two categories over a third.

For everything concerning categories and functors, while awaiting the book in preparation already mentioned, see [1] (which is necessarily very incomplete, even as regards the generalities sketched in the present no.).

Let us take this opportunity to make explicit the notion of equivalence of categories, which is not presented in a satisfactory way in [1]. A functor $F: \mathcal{C} \rightarrow \mathcal{C}'$ is said to be *faithful* (resp. *fully faithful*) if for every pair of objects S, T of $\mathcal{C}$, the map $u \mapsto F(u)$ from $\text{Hom}(S, T)$ to $\text{Hom}(F(S), F(T))$ is injective (resp. bijective). One says that F is an *equivalence* of categories if F is fully faithful, and if moreover every object S' of $\mathcal{C}'$ is isomorphic to an object of the form $F(S)$. One shows that this amounts to the same as saying that there exists a functor G from $\mathcal{C}'$ to $\mathcal{C}$ *quasi-inverse* of F , i.e., such that GF is isomorphic to $\text{id}_{\mathcal{C}}$. When this is so, the datum of a functor $G: \mathcal{C}' \rightarrow \mathcal{C}$ and of an isomorphism $\varphi: GF \rightarrow \text{id}_{\mathcal{C}}$ is equivalent to the datum, for every $S' \in \text{Ob}(\mathcal{C}')$, of a pair (S, u) formed of an object S of $\mathcal{C}$ and an isomorphism $u: F(S) \rightarrow S'$, namely $(G(S), \varphi(S))$. (With these notations, there exists a unique functor $\mathcal{C}' \rightarrow \mathcal{C}$ having the given map $S \mapsto G(S)$ as object-map, and such that the map $S \mapsto \varphi(S)$ is a homomorphism of functors $FG \rightarrow \text{id}_{\mathcal{C}'}$). Finally, if G is a functor quasi-inverse of F , and if one chooses isomorphisms $\varphi: FG \xrightarrow{\sim} \text{id}_{\mathcal{C}'}$ and $\psi: GF \xrightarrow{\sim} \text{id}_{\mathcal{C}}$, then the two compatibility conditions on φ, ψ stated in [1, I.1.2] are in fact equivalent to one another, and for every chosen isomorphism φ , there exists a unique isomorphism ψ such that the said conditions are satisfied.

2. CATEGORIES OVER ANOTHER

Let $\mathcal{E}$ be a category in $\mathbf{Univ}$, it is thus an object of $\mathbf{Cat}$, and one can consider the category $\mathbf{Cat}_{/\mathcal{E}}$ of “objects of $\mathbf{Cat}$ over $\mathcal{E}$ ”. An object of this category is thus a functor

$$p: \mathcal{F} \longrightarrow \mathcal{E}$$

One also says that the category $\mathcal{F}$, equipped with such a functor, is a *category over $\mathcal{E}$* , or an *$\mathcal{E}$ -category*. One will thus call an $\mathcal{E}$ -functor from a category $\mathcal{F}$ over $\mathcal{E}$ to a category $\mathcal{G}$ over $\mathcal{E}$, a functor

$$f: \mathcal{F} \longrightarrow \mathcal{G}$$

such that

$$qf = p$$

where p and q are the projection functors for $\mathcal{F}$ resp. $\mathcal{G}$. The set of $\mathcal{E}$ -functors f from $\mathcal{F}$ to $\mathcal{G}$ is thus in one-to-one correspondence with the set of arrows with source $\mathcal{F}$ and target $\mathcal{G}$ in $\mathbf{Cat}_{/\mathcal{E}}$, without there being an identity there, however (since the datum of an f as above does not determine $\mathcal{F}$ and $\mathcal{G}$ as categories over $\mathcal{E}$); but of course, as in any other category $\mathcal{C}_{/S}$, one will commonly make the abuse of language consisting in identifying the $\mathcal{E}$ -functors (in the sense made explicit above) with arrows in a category $\mathbf{Cat}_{/\mathcal{E}}$.

One will denote by

$$\text{Hom}_{\mathcal{E}}(\mathcal{F}, \mathcal{G})$$

the set of $\mathcal{E}$ -functors from $\mathcal{F}$ to $\mathcal{G}$. Of course, a composite of $\mathcal{E}$ -functors is an $\mathcal{E}$ -functor (the composition in question corresponding by definition to the composition of arrows in $\mathbf{Cat}_{/\mathcal{E}}$).

Consider now two $\mathcal{E}$ -functors

$$f, g: \mathcal{F} \longrightarrow \mathcal{G}$$

and a homomorphism of functors:

$$u: f \longrightarrow g$$

One says that u is an $\mathcal{E}$ -homomorphism or a “homomorphism of $\mathcal{E}$ -functors”, if for every $\xi \in \text{Ob}(\mathcal{F})$, one has

$$q(u(\xi)) = \text{id}_{p(\xi)},$$

in words: setting $S = p(\xi) = qf(\xi) = qg(\xi) \in \text{Ob}(\mathcal{E})$, the morphism

$$u(\xi): f(\xi) \longrightarrow g(\xi)$$

in $\mathcal{G}$ is an id_S -morphism. (In general, for every morphism $\alpha: T \rightarrow S$ in $\mathcal{E}$, and every category $\mathcal{G}$ over $\mathcal{E}$, a morphism v in $\mathcal{G}$ is called an α -morphism if $q(v) = \alpha$, q denoting the projection functor $\mathcal{G} \rightarrow \mathcal{E}$). If one has a third $\mathcal{E}$ -functor $h: \mathcal{F} \rightarrow \mathcal{G}$ and an $\mathcal{E}$ -homomorphism $v: g \rightarrow h$, then vu is likewise an $\mathcal{E}$ -homomorphism. Thus, the $\mathcal{E}$ -functors from $\mathcal{F}$ to $\mathcal{G}$, and the $\mathcal{E}$ -homomorphisms of such, form a subcategory of the category $\mathbf{Hom}(\mathcal{F}, \mathcal{G})$ of all functors from $\mathcal{F}$ to $\mathcal{G}$, which one will call the category of $\mathcal{E}$ -functors from $\mathcal{F}$ to $\mathcal{G}$ and which one will denote

$$\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{G})$$

It is also the kernel subcategory of the pair of functors

$$R, S: \mathbf{Hom}(\mathcal{F}, \mathcal{G}) \rightrightarrows \mathbf{Hom}(\mathcal{F}, \mathcal{E}),$$

where R is the constant functor defined by the object p of $\mathbf{Hom}(\mathcal{F}, \mathcal{E})$, and where S is the functor $f \mapsto q \circ f$ defined by $q: \mathcal{G} \rightarrow \mathcal{E}$.

To finish these generalities, it remains to define the natural pairings among the categories $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{G})$ by composition of $\mathcal{E}$ -functors. In other words, one wants to define a “composition functor”:

$$(i) \quad \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{G}) \times \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{G}, \mathcal{H}) \longrightarrow \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{H})$$

when $\mathcal{F}, \mathcal{G}, \mathcal{H}$ are three categories over $\mathcal{E}$, in such a way that this functor induce, for the objects, the composition map $(f, g) \mapsto gf$ of $\mathcal{E}$ -functors $f: \mathcal{F} \rightarrow \mathcal{G}$ and $g: \mathcal{G} \rightarrow \mathcal{H}$. For this, let us recall that one defines a canonical functor:

$$(ii) \quad \mathbf{Hom}(\mathcal{F}, \mathcal{G}) \times \mathbf{Hom}(\mathcal{G}, \mathcal{H}) \longrightarrow \mathbf{Hom}(\mathcal{F}, \mathcal{H})$$

which, for the objects, is nothing other than the composition map $(f, g) \mapsto gf$ of functors, and which transforms an arrow (u, v) , where

$$u: f \longrightarrow f', \quad v: g \longrightarrow g'$$

are arrows in $\mathbf{Hom}(\mathcal{F}, \mathcal{G})$ resp. in $\mathbf{Hom}(\mathcal{G}, \mathcal{H})$, into the arrow

$$v * u: gf \longrightarrow g'f'$$

defined by the relation:

$$v * u(\xi) = v(f'(\xi)) \cdot g(u(\xi)) = g'(u(\xi)) \cdot v(f(\xi))$$

It is well known that one thus indeed obtains a homomorphism from gf to $g'f'$, and that (for f, g and u, v variable) one thus obtains a functor (ii), i.e. that one has

$$(I) \quad \text{id}_g * \text{id}_f = \text{id}_{gf},$$

$$(II) \quad (v' * u') \circ (v * u) = (v' \circ v) * (u' \circ u)$$

Let us also recall that one has an associativity formula for the canonical pairings (ii), which is expressed on the one hand by the associativity $(hg)f = h(gf)$ of composition of functors, and on the other hand by the formula

$$(III) \quad (w * v) * u = w * (v * u)$$

for the composition products of homomorphisms of functors (where $u: f \rightarrow f'$ and $v: g \rightarrow g'$ are as above, and where one assumes given moreover a homomorphism $w: h \rightarrow h'$ of functors $h, h': \mathcal{H} \rightarrow \mathcal{K}$). I now say that *when $\mathcal{F}, \mathcal{G}$ are $\mathcal{E}$ -categories, the canonical composition functor (ii) induces a functor (i)*. As one already knows that the composite of two $\mathcal{E}$ -functors is an $\mathcal{E}$ -functor, this amounts to saying that *when $u: f \rightarrow f'$ and $v: g \rightarrow g'$ are homomorphisms of $\mathcal{E}$ -functors, then $v * u: gf \rightarrow g'f'$ is likewise a homomorphism of $\mathcal{E}$ -functors*. This follows indeed trivially from the definitions. As the pairings (i) are induced by the pairings (ii), they satisfy the same associativity property, expressed also in the formulae $(hg)f = h(gf)$ and $(w * v) * u = w * (v * u)$ for $\mathcal{E}$ -functors and $\mathcal{E}$ -homomorphisms of $\mathcal{E}$ -functors.

To complete the formulary (I), (II), (III), let us also recall the formulae:

$$(IV) \quad v * \text{id}_{\mathcal{F}} = v \quad \text{and} \quad \text{id}_{\mathcal{G}} * u = u,$$

where for simplicity one writes $v * f$ or $u * g$ instead of $v * u$, when u resp. v is the identity automorphism of f resp. g .

From the definition of the pairings (i) it follows that $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{G})$ is a functor in $\mathcal{F}, \mathcal{G}$, of the product category $\mathbf{Cat}_{/\mathcal{E}}^{\circ} \times \mathbf{Cat}_{/\mathcal{E}}$ to the category $\mathbf{Cat}$. If indeed $g: \mathcal{G} \rightarrow \mathcal{G}_1$ is an $\mathcal{E}$ -functor, i.e. an object of $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{G}, \mathcal{G}_1)$, then setting in (i) $\mathcal{H} = \mathcal{G}_1$, there corresponds to it a functor

$$g_*: \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{G}) \longrightarrow \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{G}_1)$$

One defines analogously, for an $\mathcal{E}$ -functor $f: \mathcal{F}_1 \rightarrow \mathcal{F}$, a functor

$$f^*: \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{G}) \longrightarrow \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}_1, \mathcal{G})$$

For brevity, one also denotes these functors by the symbols $f \mapsto g \circ f$ resp. $g \mapsto g \circ f$ (which designate only, in fact, the corresponding maps on the sets of objects). It follows from the associativity property indicated above that in this way, one indeed obtains as announced a functor $\mathbf{Cat}_{/\mathcal{E}}^{\circ} \times \mathbf{Cat}_{/\mathcal{E}} \rightarrow \mathbf{Cat}$.

3. CHANGE OF BASE IN CATEGORIES OVER $\mathcal{E}$

As in $\mathbf{Cat}$ the projective limits (relative to categories $\mathcal{I}$ elements of $\mathbf{Univ}$) exist, the same is true in $\mathbf{Cat}_{/\mathcal{E}}$, in particular cartesian products exist there, which are interpreted as fibered products in $\mathbf{Cat}$. In accordance with the general notations, if $\mathcal{F}$ and $\mathcal{G}$ are categories over $\mathcal{E}$, one denotes by

$$\mathcal{F} \times_{\mathcal{E}} \mathcal{G}$$

their product in $\mathbf{Cat}_{/\mathcal{E}}$, i.e. their fibered product over $\mathcal{E}$ in $\mathbf{Cat}$, as a category over $\mathcal{E}$. Thus, $\mathcal{F} \times_{\mathcal{E}} \mathcal{G}$ is equipped with two $\mathcal{E}$ -functors pr_1 and pr_2 , which define, for every category $\mathcal{H}$ over $\mathcal{E}$, a bijection

$$\text{Hom}_{\mathcal{E}}(\mathcal{H}, \mathcal{F} \times_{\mathcal{E}} \mathcal{G}) \xrightarrow{\sim} \text{Hom}_{\mathcal{E}}(\mathcal{H}, \mathcal{F}) \times \text{Hom}_{\mathcal{E}}(\mathcal{H}, \mathcal{G}).$$

This bijection arises moreover from an isomorphism of categories

$$\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{H}, \mathcal{F} \times_{\mathcal{E}} \mathcal{G}) \xrightarrow{\sim} \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{H}, \mathcal{F}) \times \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{H}, \mathcal{G})$$

by taking the sets of objects of the two sides, where the functor written is the one which has as components the functors $h \mapsto \text{pr}_1 \circ h$ and $h \mapsto \text{pr}_2 \circ h$ from the first side to the two factors of the second. The reader is left to verify that one thus indeed obtains an isomorphism (the analogous fact being true, more generally, whenever one has a projective limit of categories — and not only in the case of a fibered product).

Let us moreover recall that one has (as was said in no. 1):

$$\begin{aligned}\mathrm{Ob}(\mathcal{F} \times_{\mathcal{E}} \mathcal{G}) &= \mathrm{Ob}(\mathcal{F}) \times_{\mathrm{Ob}(\mathcal{E})} \mathrm{Ob}(\mathcal{G}) \\ \mathrm{Fl}(\mathcal{F} \times_{\mathcal{E}} \mathcal{G}) &= \mathrm{Fl}(\mathcal{F}) \times_{\mathrm{Fl}(\mathcal{E})} \mathrm{Fl}(\mathcal{G}),\end{aligned}$$

the composition of arrows being performed moreover componentwise.

In the sequel, we consider a functor

$$\lambda: \mathcal{E}' \longrightarrow \mathcal{E}$$

and for every category $\mathcal{F}$ over $\mathcal{E}$, one considers $\mathcal{F} \times_{\mathcal{E}} \mathcal{E}'$ as a category over $\mathcal{E}'$ by means of pr_2 ; in other words, we interpret the operation “fibered product” as an operation “*change of base*”, the functor $\lambda: \mathcal{E}' \rightarrow \mathcal{E}$ taking the name of “*change of base functor*”. In accordance with the well-known general facts, one thus obtains a functor, called the *change of base functor* for λ :

$$\lambda^*: \mathbf{Cat}_{/\mathcal{E}} \longrightarrow \mathbf{Cat}_{/\mathcal{E}'},$$

(adjoint of the functor “restriction of the base” which, to every category $\mathcal{F}'$ over $\mathcal{E}'$, with projection functor p' , associates $\mathcal{F}'$, regarded as a category over $\mathcal{E}$ by the functor $p = \lambda p'$). As is well known in the general case of a change of base functor in a category, the change of base functor “commutes with projective limits”, and in particular “transforms” fibered products over $\mathcal{E}$ into fibered products over $\mathcal{E}'$.

Let $\mathcal{F}$ and $\mathcal{G}$ be two categories over $\mathcal{E}$, one will define a *canonical isomorphism*:

$$(i) \quad \mathbf{Hom}_{\mathcal{E}'/-}(\mathcal{F}', \mathcal{G}') \xrightarrow{\sim} \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}', \mathcal{G})$$

$$\text{where } \mathcal{F}' = \mathcal{F} \times_{\mathcal{E}} \mathcal{E}', \mathcal{G}' = \mathcal{G} \times_{\mathcal{E}} \mathcal{E}'.$$

For this, consider the functor

$$\mathrm{pr}_1: \mathcal{G}' = \mathcal{G} \times_{\mathcal{E}} \mathcal{E}' \longrightarrow \mathcal{G},$$

and define (i) by

$$F \longmapsto \mathrm{pr}_1 \circ F,$$

which a priori designates a functor

$$(ii) \quad \mathbf{Hom}(\mathcal{F}', \mathcal{G}') \longrightarrow \mathbf{Hom}(\mathcal{F}', \mathcal{G})$$

It is therefore only necessary to verify that the latter induces a functor for the subcategories (i), and that the latter is an isomorphism. That (ii) induces a bijection

$$\mathrm{Hom}_{\mathcal{E}'/-}(\mathcal{F}', \mathcal{G}') \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{E}/-}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}', \mathcal{G})$$

is the characteristic property of the change of base functor. It remains therefore to prove that if F, G are $\mathcal{E}'$ -functors $\mathcal{F}' \rightarrow \mathcal{G}'$, then *the map*

$$u \longmapsto \mathrm{pr}_1 \circ u$$

induces a bijection

$$\mathrm{Hom}_{\mathcal{E}'}(F, G) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{E}}(\mathrm{pr}_1 \circ F, \mathrm{pr}_1 \circ G).$$

The verification of this fact is immediate, and left to the reader.

It follows from this isomorphism (i), and from the end of the preceding no., that

$$\mathbf{Hom}_{\mathcal{E}'/-}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}', \mathcal{G} \times_{\mathcal{E}} \mathcal{E}')$$

can be regarded as a functor in $\mathcal{E}', \mathcal{F}, \mathcal{G}$, from the category $\mathbf{Cat}_{/\mathcal{E}}^{\circ} \times \mathbf{Cat}_{/\mathcal{E}}^{\circ} \times \mathbf{Cat}_{/\mathcal{E}}$ to the category $\mathbf{Cat}$, isomorphic to the functor defined by the expression $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}', \mathcal{G})$. In particular, for $\mathcal{F}, \mathcal{G}$ fixed, one obtains a functor in $\mathcal{E}', \mathcal{E}' \rightarrow \mathbf{Hom}_{\mathcal{E}'/-}(\mathcal{F}', \mathcal{G}') = \mathbf{Hom}_{\mathcal{E}'/-}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}', \mathcal{G} \times_{\mathcal{E}} \mathcal{E}')$, and in particular the projection $\mathcal{E}$ -functor $\lambda: \mathcal{E}' \rightarrow \mathcal{E}$ defines a morphism i.e. a functor

$$\lambda_{\mathcal{F}, \mathcal{G}}^*: \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{G}) \longrightarrow \mathbf{Hom}_{\mathcal{E}'/-}(\mathcal{F}', \mathcal{G}')$$

which we shall make explicit. For the sets of objects of the two sides, it is the map

$$f \longmapsto f' = f \times_{\mathcal{E}} \mathcal{E}'$$

which expresses the functorial dependence of $\mathcal{F} \times_{\mathcal{E}} \mathcal{E}'$ on the object $\mathcal{F}$ over $\mathcal{E}$. On the other hand, consider two $\mathcal{E}$ -functors

$$f, g: \mathcal{F} \longrightarrow \mathcal{G}$$

and a homomorphism of $\mathcal{E}$ -functors

$$u: f \longrightarrow g,$$

we shall make explicit the corresponding homomorphism of $\mathcal{E}'$ -functors:

$$u': f' \longrightarrow g'.$$

For every

$$\xi' = (\xi, S') \in \text{Ob}(\mathcal{F}')$$

with

$$\xi \in \text{Ob}(\mathcal{F}), \quad S' \in \text{Ob}(\mathcal{E}'), \quad p(\xi) = \lambda(S') = S$$

the morphism

$$u'(\xi'): f'(\xi') = (f(\xi), S') \longrightarrow g'(\xi') = (g(\xi), S') \quad \text{in } \mathcal{G}'$$

is defined by the formula

$$u'(\xi') = (u(\xi), \text{id}_{S'})$$

(which is indeed an S' -morphism in $\mathcal{G}'$, because $q(u(\xi)) = \lambda(\text{id}_{S'}) = \text{id}_S$).

Consider now an arbitrary $\mathcal{E}$ -functor

$$\lambda': \mathcal{E}'' \longrightarrow \mathcal{E}'$$

and the corresponding functor

$$\mathbf{Hom}_{\mathcal{E}'/_-}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}', \mathcal{G} \times_{\mathcal{E}} \mathcal{E}') \longrightarrow \mathbf{Hom}_{\mathcal{E}''/_-}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}'', \mathcal{G} \times_{\mathcal{E}} \mathcal{E}''),$$

I say that this functor is nothing other than the functor one obtains by the preceding procedure, starting from $\mathcal{F}'$ and $\mathcal{G}'$ over $\mathcal{E}'$ and regarding $\mathcal{E}''$ as a category over $\mathcal{E}'$, taking into account the isomorphisms of “*transitivity of change of base*”:

$$\mathcal{F}' \times_{\mathcal{E}'} \mathcal{E}'' \xrightarrow{\sim} \mathcal{F}'' = \mathcal{F} \times_{\mathcal{E}} \mathcal{E}'' \quad \text{and} \quad \mathcal{G}' \times_{\mathcal{E}'} \mathcal{E}'' \xrightarrow{\sim} \mathcal{G}'' = \mathcal{G} \times_{\mathcal{E}} \mathcal{E}'',$$

which imply a canonical isomorphism

$$\mathbf{Hom}_{\mathcal{E}''/_-}(\mathcal{F}' \times_{\mathcal{E}'} \mathcal{E}'', \mathcal{G}' \times_{\mathcal{E}'} \mathcal{E}'') \xrightarrow{\sim} \mathbf{Hom}_{\mathcal{E}''/_-}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}'', \mathcal{G} \times_{\mathcal{E}} \mathcal{E}'')$$

The verification of this compatibility is immediate, and left to the reader.

The functors just defined are compatible with the pairings defined in the preceding no., in a precise way: if $\mathcal{F}, \mathcal{G}, \mathcal{H}$ are categories over $\mathcal{E}$ and if one sets

$$\mathcal{F}' = \mathcal{F} \times_{\mathcal{E}} \mathcal{E}', \quad \mathcal{G}' = \mathcal{G} \times_{\mathcal{E}} \mathcal{E}', \quad \mathcal{H}' = \mathcal{H} \times_{\mathcal{E}} \mathcal{E}',$$

one has commutativity in the following diagram of functors:

$$\begin{array}{ccc} \mathbf{Hom}_{\mathcal{E}'/_-}(\mathcal{F}, \mathcal{G}) \times \mathbf{Hom}_{\mathcal{E}'/_-}(\mathcal{G}, \mathcal{H}) & \longrightarrow & \mathbf{Hom}_{\mathcal{E}'/_-}(\mathcal{F}, \mathcal{H}) \\ \lambda_{\mathcal{F}, \mathcal{G}}^* \times \lambda_{\mathcal{G}, \mathcal{H}}^* \downarrow & & \downarrow \lambda_{\mathcal{F}, \mathcal{H}}^* \\ \mathbf{Hom}_{\mathcal{E}'/_-}(\mathcal{F}', \mathcal{G}') \times \mathbf{Hom}_{\mathcal{E}'/_-}(\mathcal{G}', \mathcal{H}') & \longrightarrow & \mathbf{Hom}_{\mathcal{E}'/_-}(\mathcal{F}', \mathcal{H}') \end{array}$$

where the horizontal arrows are the composition functors defined in the preceding no. This commutativity is expressed by the formulae

$$(gf)' = g'f'$$

for $f \in \text{Hom}_{\mathcal{E}}(\mathcal{F}, \mathcal{G})$, $g \in \text{Hom}_{\mathcal{E}}(\mathcal{G}, \mathcal{H})$, (a formula which simply expresses the functoriality of change of base), and the formula

$$(v * u)' = v' * u'$$

when $u: f \rightarrow f_1$ is an arrow of $\mathbf{Hom}_{\mathcal{E}'/_-}(\mathcal{F}, \mathcal{G})$ and $v: g \rightarrow g_1$ an arrow of $\mathbf{Hom}_{\mathcal{E}'/_-}(\mathcal{G}, \mathcal{H})$. The verification of this formula follows easily from the definitions.

In the sequel, we shall be interested mainly in $\mathbf{Hom}_{\mathcal{E}}(\mathcal{F}, \mathcal{G})$ (and certain remarkable subcategories of the latter) when $\mathcal{F} = \mathcal{E}$, and we introduce for this reason a special notation:

$$\Gamma(\mathcal{G}/\mathcal{E}) = \mathbf{Hom}_{\mathcal{E}}(\mathcal{E}, \mathcal{G}), \quad \Gamma(\mathcal{G}/\mathcal{E}) = \mathrm{Ob}(\Gamma(\mathcal{G}/\mathcal{E})) = \mathrm{Hom}_{\mathcal{E}}(\mathcal{E}, \mathcal{G}).$$

Remarks. When $\mathcal{E}$ is a punctual category, i.e. $\mathrm{Ob}(\mathcal{E})$ and $\mathrm{Fl}(\mathcal{E})$ reduced to a single element, which also means that $\mathcal{E}$ is a final object of the category $\mathbf{Cat}$, then the datum of a category over $\mathcal{E}$ is equivalent to the datum of just a category, (because there will be a unique functor from $\mathcal{F}$ to $\mathcal{E}$). More precisely, $\mathbf{Cat}_{/\mathcal{E}}$ is then isomorphic to $\mathbf{Cat}$. Moreover, the categories $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{G})$ are then nothing other than the $\mathbf{Hom}(\mathcal{F}, \mathcal{G})$. Let us then recall that the fundamental formula

$$\mathrm{Hom}(\mathcal{H}, \mathbf{Hom}(\mathcal{F}, \mathcal{G})) \xrightarrow{\sim} \mathrm{Hom}(\mathcal{F} \times \mathcal{H}, \mathcal{G})$$

(a functorial isomorphism in the three arguments which appear there), allows one to interpret $\mathbf{Hom}(\mathcal{F}, \mathcal{G})$ axiomatically, in terms internal to the category $\mathbf{Cat}$, so that the known formulary of the $\mathbf{Hom}$ categories appears as a special case of a formulary valid in categories such as $\mathbf{Cat}$, where “ $\mathbf{Hom}$ -objects” (defined by the preceding formula) exist. There is an analogous interpretation of $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{G})$ when one assumes once more that $\mathcal{E}$ is arbitrary, by the formula

$$\mathrm{Hom}(\mathcal{H}, \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{G})) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{E}}(\mathcal{F} \times \mathcal{H}, \mathcal{G})$$

(a functorial isomorphism in the three arguments). In this way, the formal properties set out in no. 2, 3 are special cases of more general results, valid in categories where the objects $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{G})$ (when $\mathcal{F}, \mathcal{G}$ are two objects of the category over a third $\mathcal{E}$) exist.

4. FIBER-CATEGORIES; EQUIVALENCE OF $\mathcal{E}$ -CATEGORIES

Let $\mathcal{F}$ be a category over $\mathcal{E}$, and let $S \in \mathrm{Ob}(\mathcal{E})$. One calls *fiber-category of $\mathcal{F}$ at S* the subcategory $\mathcal{F}_S$ of $\mathcal{F}$, inverse image of the punctual subcategory of $\mathcal{E}$ defined by S . Thus the objects of $\mathcal{F}_S$ are the objects ξ of $\mathcal{F}$ such that $p(\xi) = S$, its morphisms are the morphisms u of $\mathcal{F}$ such that $p(u) = \mathrm{id}_S$, i.e. the S -morphisms in $\mathcal{F}$. Of course, $\mathcal{F}_S$ is canonically isomorphic to the fibered product $\mathcal{F} \times_{\mathcal{E}} \{S\}$, where $\{S\}$ denotes the punctual subcategory of $\mathcal{E}$ defined by S , equipped with its inclusion functor into $\mathcal{E}$. It follows (taking into account the transitivity of change of base) that if one performs the change of base $\lambda: \mathcal{E}' \rightarrow \mathcal{E}$, then for every $S' \in \mathrm{Ob}(\mathcal{E}')$, the projection $\mathrm{pr}_1: \mathcal{F}' = \mathcal{F} \times_{\mathcal{E}} \mathcal{E}' \rightarrow \mathcal{F}$ induces an isomorphism

$$\mathcal{F}'_{S'} \longrightarrow \mathcal{F}_S \quad (\text{where } S = \lambda(S')).$$

Proposition 4.1. *Let $f: \mathcal{F} \rightarrow \mathcal{G}$ be an $\mathcal{E}$ -functor. If f is fully faithful, then for every change of base $\mathcal{E}' \rightarrow \mathcal{E}$, the corresponding functor $f': \mathcal{F}' = \mathcal{F} \times_{\mathcal{E}} \mathcal{E}' \rightarrow \mathcal{G}' = \mathcal{G} \times_{\mathcal{E}} \mathcal{E}'$ is fully faithful.*

The verification is immediate; more generally, one can show that every projective limit of fully faithful functors (here, f and the identity functors in $\mathcal{E}, \mathcal{E}'$) is a fully faithful functor.

One notes that the assertion analogous to 4.1, where “fully faithful” is replaced by “equivalence of categories”, is false, already for $\mathcal{G} = \mathcal{E}$. However:

Proposition 4.2. *Let $f: \mathcal{F} \rightarrow \mathcal{G}$ be an $\mathcal{E}$ -functor. The following conditions are equivalent:*

(i) *There exists an $\mathcal{E}$ -functor $g: \mathcal{G} \rightarrow \mathcal{F}$ and $\mathcal{E}$ -isomorphisms*

$$gf \xrightarrow{\sim} \mathrm{id}_{\mathcal{F}}, \quad fg \xrightarrow{\sim} \mathrm{id}_{\mathcal{G}}.$$

(ii) *For every category $\mathcal{E}'$ over $\mathcal{E}$, the functor*

$$f' = f \times_{\mathcal{E}} \mathcal{E}': \mathcal{F}' = \mathcal{F} \times_{\mathcal{E}} \mathcal{E}' \longrightarrow \mathcal{G}' = \mathcal{G} \times_{\mathcal{E}} \mathcal{E}'$$

is an equivalence of categories.

(iii) *f is an equivalence of categories, and for every $S \in \mathrm{Ob}(\mathcal{E})$, the functor $f_S: \mathcal{F}_S \rightarrow \mathcal{G}_S$ induced by f is an equivalence of categories.*

(iii bis) *f is fully faithful, and for every $S \in \mathrm{Ob}(\mathcal{E})$ and every $\eta \in \mathrm{Ob}(\mathcal{G}_S)$, there exists a $\xi \in \mathrm{Ob}(\mathcal{F}_S)$ and an S -isomorphism $u: f(\xi) \rightarrow \eta$.*

Proof. Obviously (i) implies that f is an equivalence of categories (a notion which is defined by the same condition, but where the isomorphisms of functors are not required to be $\mathcal{E}$ -morphisms). On the other hand, it follows from the functorialities of the preceding no. that condition (i) is preserved after change of base $\mathcal{E}' \rightarrow \mathcal{E}$. It follows that (i) $\Rightarrow$ (ii). Obviously (ii) $\Rightarrow$ (iii), for it suffices to take $\mathcal{E}' = \mathcal{E}$ and $\mathcal{E}' = \{S\}$. It is even more trivial that (iii) $\Rightarrow$ (iii bis), it remains to prove that (iii bis) $\Rightarrow$ (i). For this, let us choose for every $\eta \in \text{Ob}(\mathcal{G})$ a $g(\eta) \in \text{Ob}(\mathcal{F})$ and an isomorphism $u(\eta): f(g(\eta)) \rightarrow \eta$ which is such that $q(u(\eta)) = \text{id}_S$, where $S = q(\eta)$. This is possible thanks to the second condition (iii bis). As is well-known and immediate, the fact that f is fully faithful implies that g can in a unique way be regarded as a functor from $\mathcal{G}$ to $\mathcal{F}$, in such a way that the $u(\eta)$ define a functorial homomorphism (hence an isomorphism) $u: fg \xrightarrow{\sim} \text{id}_{\mathcal{G}}$. Moreover, by construction g is an $\mathcal{E}$ -functor and u an $\mathcal{E}$ -homomorphism. To the preceding data there then corresponds a functorial isomorphism $v: gf \rightarrow \text{id}_{\mathcal{F}}$, defined by the condition that $f * v = u * f$, and one sees at once that it is likewise an $\mathcal{E}$ -morphism, qed.

Definition 4.3. If the preceding conditions are satisfied, one says that f is an equivalence of categories over $\mathcal{E}$, or an $\mathcal{E}$ -equivalence.

Corollary 4.4. Suppose that the projection functor $p: \mathcal{F} \rightarrow \mathcal{E}$ is a transportable functor, i.e. that for every isomorphism $\alpha: T \rightarrow S$ in $\mathcal{E}$ and every object ξ in $\mathcal{F}_T$, there exists an isomorphism u in $\mathcal{F}$ with source ξ such that $p(u) = \alpha$. Then every $\mathcal{E}$ -functor $f: \mathcal{F} \rightarrow \mathcal{G}$ which is an equivalence of categories, is an $\mathcal{E}$ -equivalence.

Follows from the criterion (iii bis).

Corollary 4.5. Let $f: \mathcal{F} \rightarrow \mathcal{G}$ be an $\mathcal{E}$ -equivalence. Then for every category $\mathcal{H}$ over $\mathcal{E}$, the corresponding functors:

$$\begin{aligned} \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{G}, \mathcal{H}) &\longrightarrow \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{H}) \\ \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{H}, \mathcal{F}) &\longrightarrow \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{H}, \mathcal{G}) \end{aligned}$$

(cf. no. 2) are equivalences of categories.

This follows from the criterion (i) by the usual reasoning.

5. CARTESIAN MORPHISMS, INVERSE IMAGES, CARTESIAN FUNCTORS

Let $\mathcal{F}$ be a category over $\mathcal{E}$, with projection functor p .

Definition 5.1. Consider a morphism

$$\alpha: \eta \longrightarrow \xi$$

in $\mathcal{F}$, and let

$$S = p(\xi), \quad T = p(\eta), \quad f = p(\alpha).$$

One says that α is a *cartesian morphism* if for every $\eta' \in \text{Ob}(\mathcal{F}_T)$ and every f -morphism $u: \eta' \rightarrow \xi$, there exists a unique T -morphism $\bar{u}: \eta' \rightarrow \eta$ such that $u = \alpha \circ \bar{u}$.

This therefore means that for every $\eta' \in \text{Ob}(\mathcal{F}_T)$, the map $v \mapsto \alpha \circ v$:

$$(i) \quad \text{Hom}_T(\eta', \eta) \longrightarrow \text{Hom}_f(\eta', \xi)$$

is bijective. This also means that the pair (η, α) represents the functor in $\eta' \mathcal{F}_T^\circ \rightarrow \mathbf{Ens}$ of the right-hand side. If for a given morphism $f: T \rightarrow S$ in $\mathcal{E}$, and a given $\xi \in \text{Ob}(\mathcal{F}_S)$, there exists such a pair (η, α) , i.e. a cartesian morphism α in $\mathcal{F}$ with target ξ , such that $p(\alpha) = f$, then η is determined in $\mathcal{F}_T$ up to unique isomorphism. One then says that *the inverse image of ξ by f exists*, and an object η of $\mathcal{F}_T$ equipped with a cartesian f -morphism $\alpha: \eta \rightarrow \xi$ is called *an inverse image of ξ by f* . Often one assumes such an inverse image chosen whenever it exists ($\mathcal{F}$ being fixed); one then denotes the inverse image by symbols such as $f_{\mathcal{F}}^*(\xi)$, or simply $f^*(\xi)$ or $\xi \times_S T$ when these notations do not lead to confusion; the canonical morphism

$\alpha: \eta \rightarrow \xi$ will then be denoted, in what follows, by $\alpha_f(\xi)$. If for every $\xi \in \text{Ob}(\mathcal{F}_S)$, the inverse image of ξ by f exists, one will also say that *the inverse image functor by f in $\mathcal{F}$ exists*, and $f^*(\xi)$ then becomes a *functor covariant in ξ* , from $\mathcal{F}_S$ to $\mathcal{F}_T$. This comes from the fact that the right-hand side in (i) depends covariantly on ξ , i.e. more precisely designates a functor from $\mathcal{F}_T^\circ \times \mathcal{F}_S$ to **Ens**. This functorial dependence for $f^*(\xi)$ is made explicit as follows: consider cartesian f -morphisms

$$\alpha: \eta \longrightarrow \xi, \quad \alpha': \eta' \longrightarrow \xi'$$

and an S -morphism $\lambda: \xi \rightarrow \xi'$, then there exists a unique T -morphism $\mu: \eta \rightarrow \eta'$ such that one has

$$\alpha' \mu = \lambda \alpha$$

(as follows from the fact that α' is cartesian).

Let us also note the following immediate fact: consider a commutative diagram

$$\begin{array}{ccc} \xi & \xleftarrow{\alpha} & \eta \\ \lambda \downarrow & & \downarrow \mu \\ \xi' & \xleftarrow{\alpha'} & \eta' \end{array}$$

in $\mathcal{F}$, where α and α' are f -morphisms, and λ an S -isomorphism, μ a T -isomorphism. For α to be cartesian, it is necessary and sufficient that α' be so.

Definition 5.2. An $\mathcal{E}$ -functor $F: \mathcal{F} \rightarrow \mathcal{G}$ is called a *cartesian functor* if it transforms cartesian morphisms into cartesian morphisms. One denotes by $\mathbf{Hom}_{\text{cart}}(\mathcal{F}, \mathcal{G})$ the full subcategory of $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{G})$ formed of the cartesian functors.

For example, considering $\mathcal{E}$ as a category over $\mathcal{E}$ by means of the identity functor, every morphism of $\mathcal{E}$ is cartesian, hence a cartesian functor from $\mathcal{E}$ to $\mathcal{F}$ is a section functor $F: \mathcal{E} \rightarrow \mathcal{F}$ which transforms every morphism of $\mathcal{E}$ into a cartesian morphism; such a functor is called a *cartesian section* of $\mathcal{F}$ over $\mathcal{E}$.

Proposition 5.3. (i) *A functor $F: \mathcal{F} \rightarrow \mathcal{G}$ which is an $\mathcal{E}$ -equivalence, is a cartesian functor.*
(ii) *Let F, G be two isomorphic $\mathcal{E}$ -functors $\mathcal{F} \rightarrow \mathcal{G}$. If one is cartesian, so is the other.*
(iii) *The composite of two cartesian functors $\mathcal{F} \rightarrow \mathcal{G}$ and $\mathcal{G} \rightarrow \mathcal{H}$ is a cartesian functor.*

Assertion (iii) is trivial from the definition, (ii) follows from the remark preceding 5.2, (i) follows easily from the definition and from the criterion 4.2 (iii); more precisely, a morphism α in $\mathcal{F}$ is cartesian if and only if $F(\alpha)$ is so.

Corollary 5.4. *Let $F: \mathcal{F} \rightarrow \mathcal{G}$ be an $\mathcal{E}$ -equivalence. Then for every category $\mathcal{H}$ over $\mathcal{E}$, the corresponding functors $G \mapsto G \circ F$ and $G \mapsto F \circ G$ induce equivalences of categories:*

$$\begin{aligned} \mathbf{Hom}_{\text{cart}}(\mathcal{G}, \mathcal{H}) &\xrightarrow{\approx} \mathbf{Hom}_{\text{cart}}(\mathcal{F}, \mathcal{H}) \\ \mathbf{Hom}_{\text{cart}}(\mathcal{H}, \mathcal{F}) &\xrightarrow{\approx} \mathbf{Hom}_{\text{cart}}(\mathcal{H}, \mathcal{G}) \end{aligned}$$

This is deduced in the usual way from 4.2 criterion (i) and from 5.3 (i) (ii) (iii). One can make more precise that *the $\mathcal{E}$ -functor $G: \mathcal{G} \rightarrow \mathcal{H}$ is cartesian if and only if $G \circ F$ is so*, and likewise *an $\mathcal{E}$ -functor $G: \mathcal{H} \rightarrow \mathcal{F}$ is cartesian if and only if $F \circ G$ is so*.

It follows from 5.4 (iii) that if one considers the subcategory $\mathbf{Cat}_{/\mathcal{E}}^{\text{cart}}$ of $\mathbf{Cat}_{/\mathcal{E}}$ whose objects are the same as those of $\mathbf{Cat}_{/\mathcal{E}}$, and whose morphisms are the *cartesian* functors, then one has as in no. 2 pairings:

$$\mathbf{Hom}_{\text{cart}}(\mathcal{F}, \mathcal{G}) \times \mathbf{Hom}_{\text{cart}}(\mathcal{G}, \mathcal{H}) \longrightarrow \mathbf{Hom}_{\text{cart}}(\mathcal{F}, \mathcal{H})$$

induced by those of no. 2, allowing one to consider $\mathbf{Hom}_{\text{cart}}(\mathcal{F}, \mathcal{G})$ as a functor in $\mathcal{F}, \mathcal{G}$, from the category $\left(\mathbf{Cat}_{/\mathcal{E}}^{\text{cart}}\right)^\circ \times \mathbf{Cat}_{/\mathcal{E}}^{\text{cart}}$ to **Cat**. We will need this remark especially for the case where $\mathcal{F} = \mathcal{G}$:

Definition 5.5. Let $\mathcal{F}$ be a category over $\mathcal{E}$. One denotes by

$$\varprojlim \mathcal{F}/\mathcal{E}$$

the category of cartesian $\mathcal{E}$ -functors $\mathcal{E} \rightarrow \mathcal{F}$, i.e. of cartesian sections of $\mathcal{F}$ over $\mathcal{E}$.

According to what has just been said, $\varprojlim \mathcal{F}/\mathcal{E}$ is a functor in $\mathcal{F}$, from the category $\mathbf{Cat}_{/\mathcal{E}}^{\text{cart}}$ to the category $\mathbf{Cat}$.

We will see below the relations between this operation $\varprojlim$ and the notion of projective limit of categories, as well as numerous examples.

6. FIBERED CATEGORIES AND PREFIBERED CATEGORIES. PRODUCTS AND CHANGE OF BASE THEREIN

Definition 6.1. A category $\mathcal{F}$ over $\mathcal{E}$ is called a *fibered category* (and one then says that the functor $\mathcal{F} \rightarrow \mathcal{E}$ is *fibrant*) if it satisfies the following two axioms:

- Fib_I For every morphism $f: T \rightarrow S$ in $\mathcal{E}$, the inverse image functor by f in $\mathcal{F}$ exists.
- Fib_{II} The composite of two cartesian morphisms is cartesian.

A category $\mathcal{F}$ over $\mathcal{E}$ satisfying the condition Fib_I is called a *prefibered category over $\mathcal{E}$* .

If $\mathcal{F}$ is a fibered category (resp. prefibered) over $\mathcal{E}$, a subcategory $\mathcal{G}$ of $\mathcal{F}$ is called a *fibered subcategory* (resp. a *prefibered subcategory*) if it is a fibered category (resp. prefibered) over $\mathcal{E}$, and if moreover the inclusion functor is cartesian. If for example $\mathcal{G}$ is a *full* subcategory of $\mathcal{F}$, one sees that this means that for every morphism $f: T \rightarrow S$ in $\mathcal{E}$ and for every $\xi \in \text{Ob}(\mathcal{G}_S)$, $f_{\mathcal{F}}^*(\xi)$ is T -isomorphic to an object of $\mathcal{G}_T$. Another interesting case is the following: $\mathcal{F}$ being a fibered category over $\mathcal{E}$, consider the subcategory $\mathcal{G}$ of $\mathcal{F}$ having the same objects, and whose morphisms are the *cartesian* morphisms of $\mathcal{F}$; in particular the morphisms of $\mathcal{G}_S$ are the isomorphisms of $\mathcal{F}_S$. One sees at once that this is indeed a fibered subcategory of $\mathcal{F}$, for in the bijection

$$\text{Hom}_T(\eta', \eta) \xrightarrow{\sim} \text{Hom}_f(\eta', \xi)$$

relative to a cartesian f -morphism α in $\mathcal{F}$, to the T -isomorphisms of the left-hand side correspond the cartesian morphisms of the right-hand side. By definition, the cartesian sections $\mathcal{E} \rightarrow \mathcal{F}$ then correspond bijectively to arbitrary $\mathcal{E}$ -functors $\mathcal{E} \rightarrow \mathcal{G}$ (but one notes that the natural functor

$$\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{E}, \mathcal{G}) \longrightarrow \mathbf{Hom}_{\text{cart}}(\mathcal{E}, \mathcal{F}) = \varprojlim(\mathcal{F}/\mathcal{E})$$

is faithful, but in general is not fully faithful, i.e. is not an isomorphism).

Remarks. Let $\mathcal{F}$ be a category over $\mathcal{E}$. The following conditions are equivalent: (i) All morphisms of $\mathcal{F}$ are cartesian (ii) $\mathcal{F}$ is a fibered category over $\mathcal{E}$, and the $\mathcal{F}_S$ are groupoids (i.e. every morphism in $\mathcal{F}_S$ is an isomorphism). One then says that $\mathcal{F}$ is a category *fibered in groupoids* over $\mathcal{E}$. These are the ones one encounters especially in “theory of modules”. If $\mathcal{E}$ is a groupoid, one shows that conditions (i) and (ii) are also equivalent to the following: (iii) $\mathcal{F}$ is a groupoid, and the projection functor $p: \mathcal{F} \rightarrow \mathcal{E}$ is transportable (cf. 4.4). For example, if $\mathcal{E}$ and $\mathcal{F}$ are groupoids such that $\text{Ob}(\mathcal{E})$ and $\text{Ob}(\mathcal{F})$ are reduced to a point, so that $\mathcal{E}$ and $\mathcal{F}$ are defined, up to isomorphism, by groups E and F , and the functor $p: \mathcal{F} \rightarrow \mathcal{E}$ is defined by a group homomorphism $p: F \rightarrow E$, then $\mathcal{F}$ is fibered over $\mathcal{E}$ if and only if p is surjective, i.e. if p defines an *extension* of the group E by the group $G = \text{Ker } p$.

Proposition 6.2. *Let $F: \mathcal{F} \rightarrow \mathcal{G}$ be an $\mathcal{E}$ -equivalence. For $\mathcal{F}$ to be a fibered category (resp. prefibered) over $\mathcal{E}$, it is necessary and sufficient that $\mathcal{G}$ be so.*

Follows easily from the definitions and from the remark noted above that a morphism α in $\mathcal{F}$ is cartesian if and only if $F(\alpha)$ is so.

Proposition 6.3. *Let $\mathcal{F}_1, \mathcal{F}_2$ be two categories over $\mathcal{E}$, and let $\alpha = (\alpha_1, \alpha_2)$ be a morphism in $\mathcal{F} = \mathcal{F}_1 \times_{\mathcal{E}} \mathcal{F}_2$. For α to be cartesian, it is necessary and sufficient that its components be so.*

Indeed, let ξ_i be the target and η_i the source of α_i , and let $f: T \rightarrow S$ be the morphism of $\mathcal{E}$ such that α_1 and α_2 are f -morphisms. For every $\eta' = (\eta'_1, \eta'_2)$ in $\mathcal{F}_T$, one has a commutative diagram

$$\begin{array}{ccc} \mathrm{Hom}_T(\eta', \eta) & \xrightarrow{\quad\quad\quad} & \mathrm{Hom}_f(\eta', \xi) \\ \downarrow & & \downarrow \\ \mathrm{Hom}_T(\eta'_1, \eta_1) \times \mathrm{Hom}_T(\eta'_2, \eta_2) & \xrightarrow{\quad\quad\quad} & \mathrm{Hom}_f(\eta'_1, \xi_1) \times \mathrm{Hom}_f(\eta'_2, \xi_2) \end{array}$$

where the vertical arrows are bijections. Thus if one of the horizontal arrows is a bijection, so is the other. This already shows that if α_1, α_2 are cartesian (hence the second horizontal arrow bijective) then α is so. The converse is seen by setting in the diagram above $\eta'_i = \eta_i$ whence $\mathrm{Hom}_T(\eta'_i, \eta_i) \neq \emptyset$, first for $i = 2$ which proves that α_1 is cartesian, then for $i = 1$ which proves that α_2 is cartesian.

Corollary 6.4. *Let $\mathcal{F} = \mathcal{F}_1 \times_{\mathcal{E}} \mathcal{F}_2$, and let $F = (F_1, F_2)$ be an $\mathcal{E}$ -functor $\mathcal{G} \rightarrow \mathcal{F}$. For F to be cartesian, it is necessary and sufficient that F_1 and F_2 be so. One thus obtains an isomorphism of categories*

$$\mathbf{Hom}_{\mathrm{cart}}(\mathcal{G}, \mathcal{F}_1 \times_{\mathcal{E}} \mathcal{F}_2) \xrightarrow{\sim} \mathbf{Hom}_{\mathrm{cart}}(\mathcal{G}, \mathcal{F}_1) \times \mathbf{Hom}_{\mathrm{cart}}(\mathcal{G}, \mathcal{F}_2)$$

and in particular (setting $\mathcal{G} = \mathcal{E}$) an isomorphism of categories

$$\varprojlim(\mathcal{F}_1 \times_{\mathcal{E}} \mathcal{F}_2 / \mathcal{E}) \xrightarrow{\sim} \varprojlim(\mathcal{F}_1 / \mathcal{E}) \times \varprojlim(\mathcal{F}_2 / \mathcal{E})$$

Corollary 6.5. *Let $\mathcal{F}_1$ and $\mathcal{F}_2$ be two fibered categories (resp. prefibered) over $\mathcal{E}$, then their fibered product $\mathcal{F} = \mathcal{F}_1 \times_{\mathcal{E}} \mathcal{F}_2$ is a fibered category (resp. prefibered) over $\mathcal{E}$.*

These results moreover extend to the case of the fibered product of an arbitrary family of categories over $\mathcal{E}$.

Proposition 6.6. *Let $\mathcal{F}$ be a category over $\mathcal{E}$, with projection functor p , and let $\lambda: \mathcal{E}' \rightarrow \mathcal{E}$ be a functor, consider $\mathcal{F}' = \mathcal{F} \times_{\mathcal{E}} \mathcal{E}'$ as a category over $\mathcal{E}'$ by the projection functor $p' = p \times_{\mathcal{E}} \mathrm{id}_{\mathcal{E}'}$. Let α' be a morphism of $\mathcal{F}'$; for α' to be a cartesian morphism, it is necessary and sufficient that its image α in $\mathcal{F}$ be so.*

The proof is immediate and left to the reader.

Corollary 6.7. *For every cartesian functor $F: \mathcal{F} \rightarrow \mathcal{G}$ of categories over $\mathcal{E}$, the functor $F' = F \times_{\mathcal{E}} \mathcal{E}'$ from $\mathcal{F}' = \mathcal{F} \times_{\mathcal{E}} \mathcal{E}'$ to $\mathcal{G}' = \mathcal{G} \times_{\mathcal{E}} \mathcal{E}'$ is cartesian.*

Consequently, the functor $\mathbf{Hom}_{\mathcal{E}}(\mathcal{F}, \mathcal{G}) \rightarrow \mathbf{Hom}_{\mathcal{E}'}(\mathcal{F}', \mathcal{G}')$ considered in no. 3 induces a functor

$$\mathbf{Hom}_{\mathrm{cart}}(\mathcal{F}, \mathcal{G}) \longrightarrow \mathbf{Hom}_{\mathrm{cart}}(\mathcal{F}', \mathcal{G}');$$

in other terms, for $\mathcal{F}, \mathcal{G}$ fixed, one can consider

$$\mathbf{Hom}_{\mathrm{cart}}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}', \mathcal{G} \times_{\mathcal{E}} \mathcal{E}')$$

as a functor in $\mathcal{E}'$, from the category $\mathbf{Cat}_{/\mathcal{E}}^{\circ}$ to $\mathbf{Cat}$. If one likewise lets $\mathcal{F}, \mathcal{G}$ vary, one finds a functor from the category $\mathbf{Cat}_{/\mathcal{E}}^{\circ} \times (\mathbf{Cat}_{/\mathcal{E}}^{\mathrm{cart}})^{\circ} \times \mathbf{Cat}_{/\mathcal{E}}^{\mathrm{cart}}$ to $\mathbf{Cat}$. When one takes into account the isomorphism

$$\mathrm{Hom}_{\mathcal{E}'}(\mathcal{F}', \mathcal{G}') \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{E}}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}', \mathcal{G})$$

envisaged in no. 3, then the cartesian $\mathcal{E}'$ -functors from $\mathcal{F}'$ to $\mathcal{G}'$ correspond to the $\mathcal{E}$ -functors $\mathcal{F} \times_{\mathcal{E}} \mathcal{E}' \rightarrow \mathcal{G}$ which transform every morphism whose first projection is a cartesian morphism of $\mathcal{F}$, into a cartesian morphism of $\mathcal{G}$. Setting $\mathcal{F} = \mathcal{E}$, one finds (after change of notation):

Corollary 6.8. *$\varprojlim(\mathcal{F}' / \mathcal{E}')$ is isomorphic to the full subcategory of $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{E}', \mathcal{F})$ formed of the $\mathcal{E}$ -functors $\mathcal{E}' \rightarrow \mathcal{F}$ which transform arbitrary morphisms into cartesian morphisms. In particular, if $\mathcal{F}$ is a fibered category and if $\tilde{\mathcal{F}}$ is the subcategory of $\mathcal{F}$ whose morphisms are the cartesian morphisms of $\mathcal{F}$, then one has a bijection*

$$\mathrm{Ob} \varprojlim(\mathcal{F}' / \mathcal{E}') \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{E}/-}(\mathcal{E}', \tilde{\mathcal{F}}).$$

This makes more precise the way in which the expression $\varprojlim(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}' / \mathcal{E}')$ is to be considered as a functor in $\mathcal{E}'$ and in $\mathcal{F}$, from the category $\mathbf{Cat}_{/\mathcal{E}^\circ} \times \mathbf{Cat}_{/\mathcal{E}}^{\text{cart}}$ to the category $\mathbf{Cat}$. One will see later a more complete functorial dependence with respect to $\mathcal{E}'$, when $\mathcal{F}$ is required to be a fibered category.

Corollary 6.9. *Let $\mathcal{F}$ be a fibered category (resp. prefibered) over $\mathcal{E}$, then $\mathcal{F}' = \mathcal{F} \times_{\mathcal{E}} \mathcal{E}'$ is a fibered category (resp. prefibered) over $\mathcal{E}'$.*

Proposition 6.10. *Let $\mathcal{F}, \mathcal{G}$ be prefibered categories over $\mathcal{E}$, F a cartesian $\mathcal{E}$ -functor from $\mathcal{F}$ to $\mathcal{G}$. For F to be faithful, (resp. fully faithful, resp. an $\mathcal{E}$ -equivalence) it is necessary and sufficient that for every $S \in \text{Ob}(\mathcal{E})$, the induced functor $F_S: \mathcal{F}_S \rightarrow \mathcal{G}_S$ be faithful (resp. fully faithful, resp. an equivalence).*

Immediate proof from the definitions.

To finish this no., we give some properties of fibered categories, using the axiom Fib_{II} .

Proposition 6.11. *Let $\mathcal{F}$ be a prefibered category over $\mathcal{E}$. For $\mathcal{F}$ to be fibered, it is necessary and sufficient that it satisfy the following condition:*

Fib'_{II} : *Let $\alpha: \eta \rightarrow \xi$ be a cartesian morphism in $\mathcal{F}$ over the morphism $f: T \rightarrow S$ of $\mathcal{E}$. For every morphism $g: U \rightarrow T$ in $\mathcal{E}$, and every $\zeta \in \text{Ob}(\mathcal{F}_U)$, the map $u \mapsto \alpha \circ u$:*

$$\text{Hom}_g(\zeta, \eta) \longrightarrow \text{Hom}_{fg}(\zeta, \xi)$$

is bijective.

In other terms, in a fibered category over $\mathcal{E}$, cartesian diagrams are characterized by a property, a priori stronger than that of the definition (which one obtains by setting $g = \text{id}_T$ in the preceding statement).

Corollary 6.12. *Let $\mathcal{F}$ be a category over $\mathcal{E}$, α a morphism in $\mathcal{F}$. For α to be an isomorphism, it is necessary that $p(\alpha) = f$ be an isomorphism and that α be cartesian; the converse is true if $\mathcal{F}$ is fibered over $\mathcal{E}$.*

Indeed, if α is an isomorphism then so is evidently $f = p(\alpha)$; for every $\eta' \in \text{Ob}(\mathcal{F}_T)$, the map $u \mapsto \alpha \circ u$

$$\text{Hom}(\eta', \eta) \longrightarrow \text{Hom}(\eta', \xi)$$

is bijective; as f is an isomorphism, one sees at once that an element of the left-hand side is a T -morphism if and only if its image in the right-hand side is an f -morphism, hence one thus obtains a bijection

$$\text{Hom}_T(\eta', \eta) \longrightarrow \text{Hom}_f(\eta', \xi)$$

which proves the first assertion. Conversely, suppose that f is an isomorphism and that α satisfies the condition stated in Fib'_{II} (which therefore means, when $\mathcal{F}$ is fibered over $\mathcal{E}$, that α is cartesian), then one sees at once that for every $\zeta \in \text{Ob} \mathcal{F}$, the map $u \mapsto \alpha \circ u$ from $\text{Hom}(\zeta, \eta)$ to $\text{Hom}(\zeta, \xi)$ is bijective, hence α is an isomorphism.

Corollary 6.13. *Let $\alpha: \eta \rightarrow \xi$ and $\beta: \zeta \rightarrow \eta$ be two composable morphisms in the category $\mathcal{F}$ fibered over $\mathcal{E}$. If α is cartesian then β is so if and only if $\alpha\beta$ is so.*

One uses the definition of cartesian morphisms in the strengthened form of 6.11.

7. CLOVEN CATEGORIES OVER $\mathcal{E}$

Definition 7.1. Let $\mathcal{F}$ be a category over $\mathcal{E}$. One calls *cleavage* of $\mathcal{F}$ over $\mathcal{E}$ a function which attaches to every $f \in \text{Fl}(\mathcal{E})$ an inverse image functor for f in $\mathcal{F}$, namely f^* . The cleavage is said to be *normalized* if $f = \text{id}_S$ implies $f^* = \text{id}_{\mathcal{F}_S}$. One calls *cloven category* (resp. *normalized cloven category*) a category $\mathcal{F}$ over $\mathcal{E}$ equipped with a cleavage (resp. with a normalized cleavage).

It is clear that $\mathcal{F}$ admits a cleavage if and only if $\mathcal{F}$ is prefibered over $\mathcal{E}$, and then $\mathcal{F}$ admits a normalized cleavage. The set of cleavages on $\mathcal{F}$ is in bijective correspondence with the set of subsets K of $\text{Fl}(\mathcal{F})$ satisfying the following conditions:

- a) The $\alpha \in K$ are cartesian morphisms.
- b) For every morphism $f: T \rightarrow S$ in $\mathcal{E}$ and every $\xi \in \text{Ob}(\mathcal{F}_S)$, there exists a unique f -morphism in K , with target ξ .

For the cleavage defined by K to be normalized, it is necessary and sufficient that K satisfy in addition the condition

- c) The identity morphisms in $\mathcal{F}$ belong to K .

The morphisms that are elements of K may be called the “*transport morphisms*” for the cleavage under consideration.

The notion of isomorphism of cloven categories over $\mathcal{E}$ is clear. More generally, one may define the morphisms of cloven $\mathcal{E}$ -categories as the functors of $\mathcal{E}$ -categories $\mathcal{F} \rightarrow \mathcal{G}$ which send transport morphisms to transport morphisms. (These are in particular cartesian functors). In this way the cloven categories over $\mathcal{E}$ are the objects of a category, the *category of cloven categories over $\mathcal{E}$* . The reader will spell out the existence of products, related to the fact that if a category over $\mathcal{E}$ is a product of categories $\mathcal{F}_i$ over $\mathcal{E}$ each equipped with a cleavage, then $\mathcal{F}$ is equipped with a corresponding natural cleavage. One also leaves to the reader to spell out the notion of change of base in cloven categories.

We shall denote by $\alpha_f(\xi)$ the canonical morphism

$$\alpha_f(\xi): f^*(\xi) \longrightarrow \xi.$$

It is, as has been said, functorial in ξ , i.e. one has a functorial homomorphism

$$\alpha_f: i_T f^* \longrightarrow i_S,$$

where for every $S \in \text{Ob}(\mathcal{E})$, i_S denotes the inclusion functor

$$i_S: \mathcal{F}_S \longrightarrow \mathcal{F}$$

Consider now morphisms

$$f: T \longrightarrow S \quad \text{and} \quad g: U \longrightarrow T$$

in $\mathcal{E}$, and let $\xi \in \text{Ob}(\mathcal{F}_S)$, there then exists a unique U -morphism

$$c_{f,g}(\xi): g^* f^*(\xi) \longrightarrow (fg)^*(\xi)$$

making the diagram commutative

$$\begin{array}{ccc} f^*(\xi) & \xleftarrow{\alpha_g(f^*(\xi))} & g^*(f^*(\xi)) \\ \alpha_f(\xi) \downarrow & & \downarrow c_{f,g}(\xi) \\ \xi & \xleftarrow{\alpha_{fg}(\xi)} & (fg)^*(\xi) \end{array}$$

(by virtue of the definition of $(fg)^*(\xi)$). For variable ξ , this homomorphism is functorial, i.e.: one has a homomorphism

$$c_{f,g}: g^* f^* \longrightarrow (fg)^*$$

of functors $\mathcal{F}_S \rightarrow \mathcal{F}_U$. Let us note at once:

Proposition 7.2. *For the cloven category $\mathcal{F}$ over $\mathcal{E}$ to be fibered, it is necessary and sufficient that the $c_{f,g}$ be isomorphisms.*

One concludes from this, taking for f an isomorphism, for g its inverse, and considering the isomorphisms $c_{f,g}$ and $c_{g,f}$:

Corollary 7.3. *If $\mathcal{F}$ is a fibered cloven category over $\mathcal{E}$, then for every isomorphism $f: T \rightarrow S$ in $\mathcal{E}$, f^* is an equivalence of categories $\mathcal{F}_S \rightarrow \mathcal{F}_T$.*

Proposition 7.4. *Let $\mathcal{F}$ be a cloven category over $\mathcal{E}$. One has*

$$\begin{aligned} A) \quad & \begin{cases} c_{f, \text{id}_T}(\xi) = \alpha_{\text{id}_T}(f^*(\xi)) \\ c_{\text{id}_S, f}(\xi) = f^*(\alpha_{\text{id}_S}(\xi)) \end{cases} \\ B) \quad & c_{f, gh}(\xi) \cdot c_{g, h}(f^*(\xi)) = c_{fg, h}(\xi) \cdot h^*(c_{f, g}(\xi)) \end{aligned}$$

(In these formulas, f, g, h denote morphisms

$$V \longrightarrow U \longrightarrow T \longrightarrow S$$

and ξ an object of $\mathcal{F}_S$).

The first and second relations, in the case of a normalized cleavage, take the simpler form

$$A') \quad c_{f, \text{id}_T} = \text{id}_{f^*}, \quad c_{\text{id}_S, f} = \text{id}_{f^*}.$$

As for the third, it is visualized by the commutativity of the diagram

$$(D) \quad \begin{array}{ccc} h^* g^* f^*(\xi) & \xrightarrow{c_{g, h}(f^*(\xi))} & (gh)^*(f^*(\xi)) \\ h^*(c_{f, g}(\xi)) \downarrow & & \downarrow c_{f, gh}(\xi) \\ h^*(fg)^*(\xi) & \xrightarrow{c_{fg, h}(\xi)} & (fgh)^*(\xi) \end{array}$$

In the case of fibered categories (where the $c_{f, g}$ are isomorphisms), this commutativity can be expressed intuitively by the fact that *the successive use of isomorphisms of the form $c_{f, g}$ does not lead to “contradictory identifications”*. One can equally write this formula without argument ξ , by using the convolution product of homomorphisms of functors:

$$c_{fg, h} \circ (h^* * c_{f, g}) = c_{f, gh} \circ (c_{g, h} * f^*).$$

The proof of the first two formulas 7.4 is trivial; let us sketch that of the third. For this, consider, in addition to the square (D), the square of homomorphisms:

$$(D') \quad \begin{array}{ccc} g^* f^*(\xi) & \xrightarrow{\alpha_g(f^*(\xi))} & f^*(\xi) \\ c_{f, g}(\xi) \downarrow & & \downarrow \alpha_f(\xi) \\ (fg)^*(\xi) & \xrightarrow{\alpha_{fg}(\xi)} & \xi \end{array}$$

which is commutative by definition of $c_{f, g}(\xi)$. Consider the diagram obtained by joining the vertices of (D) to the corresponding vertices of (D') by the homomorphisms of the form α :

$$\begin{aligned} & \alpha_h(g^* f^*(\xi)), \quad \alpha_{gh}(f^*(\xi)) \\ & \alpha_h((fg)^*(\xi)), \quad \alpha_{fgh}(\xi). \end{aligned}$$

The four lateral faces of the cube thus obtained are likewise commutative: for the left-hand one, this comes from the fact that the left-hand column of (D) is deduced from the left-hand column of (D') by applying h , and that α_h is a functorial homomorphism; for the other three, it is nothing other than the definition of the operations c of the three remaining sides of (D). Thus the five faces of the cube other than the top face are commutative. It follows that the two (fgh) -morphisms $h^* g^* f^*(\xi) \rightarrow (fgh)^*(\xi)$ defined by (D) have the same composite with $\alpha_{fgh}(\xi): (fgh)^*(\xi) \rightarrow \xi$, hence they are equal by definition of $(fgh)^*$.

Let us restrict ourselves in what follows to *normalized* cloven categories. Such a category gives rise to the following objects:

- a) A map $S \mapsto \mathcal{F}_S$ from $\text{Ob}(\mathcal{E})$ to $\mathbf{Cat}$.
- b) A map $f \mapsto f^*$, associating to every $f \in \text{Fl}(\mathcal{E})$, with source T and target S , a functor $f^*: \mathcal{F}_S \rightarrow \mathcal{F}_T$.
- c) A map $(f, g) \mapsto c_{f, g}$, associating to every pair of arrows (f, g) of $\mathcal{E}$, a functorial homomorphism $c_{f, g}: g^* f^* \rightarrow (fg)^*$.

Moreover these data satisfy the conditions expressed in the formulas $A')$ and $B)$ given above. (N.B. If one had not restricted oneself to the case of a normalized cleavage, it would have been necessary to introduce an additional object, namely a function $S \mapsto \alpha_S$ which associates to every object S of $\mathcal{E}$ a functorial homomorphism $\alpha_S: (\text{id}_S)^* \rightarrow \text{id}_{\mathcal{F}_S}$; the condition $A')$ would then be replaced by the condition $A)$).

We shall now show how one can reconstruct (up to unique isomorphism) the normalized cloven category $\mathcal{F}$ over $\mathcal{E}$ with the help of the preceding objects.

8. CLOVEN CATEGORY DEFINED BY A PSEUDOFUNCTOR $\mathcal{E}^\circ \rightarrow \mathbf{Cat}$

Let us call, for brevity, a *pseudofunctor* from $\mathcal{E}^\circ$ to $\mathbf{Cat}$ (one should say, a *normalized pseudofunctor*), a set of data a), b), c) as above, satisfying the conditions $A')$ and $B)$. In the preceding no., we associated, to a normalized cloven category over $\mathcal{E}$, a pseudofunctor $\mathcal{E}^\circ \rightarrow \mathbf{Cat}$; here we shall indicate the inverse construction. We shall leave to the reader the verification of most of the details, as well as of the fact that these constructions are indeed “inverse” to one another. More precisely, one should consider the pseudofunctors $\mathcal{E}^\circ \rightarrow \mathbf{Cat}$ as the objects of a new category, and show that our constructions furnish equivalences, quasi-inverse to one another, between the latter and the category of cloven categories over $\mathcal{E}$, defined in the preceding no.

One sets

$$\mathcal{F}_\circ = \coprod_{S \in \text{Ob}(\mathcal{E})} \text{Ob}(\mathcal{F}(S)),$$

the sum of the sets $\text{Ob}(\mathcal{F}(S))$ (N.B. we shall write here $\mathcal{F}(S)$ and not $\mathcal{F}_S$ for the value at the object S of $\mathcal{E}$ of the given pseudofunctor, to avoid confusions of notation in what follows). One thus has an obvious map:

$$p_\circ: \mathcal{F}_\circ \longrightarrow \text{Ob } \mathcal{E}.$$

Let

$$\bar{\xi} = (S, \xi), \quad \bar{\eta} = (T, \eta) \quad (\text{with } \xi \in \text{Ob } \mathcal{F}(S), \eta \in \text{Ob } \mathcal{F}(T))$$

be two elements of $\mathcal{F}_\circ$, and let $f \in \text{Hom}(T, S)$, one will set

$$h_f(\bar{\eta}, \bar{\xi}) = \text{Hom}_{\mathcal{F}(T)}(\eta, f^*(\xi)).$$

If moreover one has a morphism $g: U \rightarrow T$ in $\mathcal{E}$, and a $\zeta \in \text{Ob } \mathcal{F}(U)$, one defines a map, denoted $(u, v) \mapsto u \circ v$:

$$h_f(\bar{\eta}, \bar{\xi}) \times h_g(\bar{\zeta}, \bar{\eta}) \longrightarrow h_{fg}(\bar{\zeta}, \bar{\xi}),$$

i.e. a map

$$\text{Hom}_{\mathcal{F}(T)}(\eta, f^*(\xi)) \times \text{Hom}_{\mathcal{F}(U)}(\zeta, g^*(\eta)) \longrightarrow \text{Hom}_{\mathcal{F}(U)}(\zeta, (fg)^*(\xi)),$$

by the formula

$$u \circ v = c_{f,g}(\xi) \cdot g^*(u) \cdot v,$$

i.e. $u \circ v$ is the composite of the sequence

$$\zeta \xrightarrow{u} g^*(\eta) \xrightarrow{g^*(u)} g^*f^*(\xi) \xrightarrow{c_{f,g}(\xi)} (fg)^*(\xi).$$

On the other hand one will set

$$h(\bar{\eta}, \bar{\xi}) = \coprod_{f \in \text{Hom}(T, S)} h_f(\bar{\eta}, \bar{\xi}),$$

and the preceding pairings define pairings

$$h(\bar{\eta}, \bar{\xi}) \times h(\bar{\zeta}, \bar{\eta}) \longrightarrow h(\bar{\zeta}, \bar{\xi}),$$

while the definition of the $h(\bar{\eta}, \bar{\xi})$ implies an obvious map:

$$p_{\bar{\eta}, \bar{\xi}}: h(\bar{\eta}, \bar{\xi}) \longrightarrow \text{Hom}(T, S).$$

This being said, one verifies the following points:

- 1) The composition between elements of the $h(\bar{\eta}, \bar{\xi})$ is *associative*.

- 2) For every $\bar{\xi} = (\xi, S)$ in $\mathcal{F}_o$, consider the identity element of

$$h_{\text{id}_S}(\bar{\xi}, \bar{\xi}) = \text{Hom}_{\mathcal{F}(S)}(\text{id}_S^*(\xi), \xi) = \text{Hom}_{\mathcal{F}(S)}(\xi, \xi),$$

and its image in $h(\bar{\xi}, \bar{\xi})$. This object is a *unit* on the left and on the right for the composition between elements of the $h(\bar{\eta}, \bar{\xi})$.

This already shows that *one obtains a category* $\mathcal{F}$, by setting

$$\text{Ob } \mathcal{F} = \mathcal{F}_o, \quad \text{Fl } \mathcal{F} = \coprod_{\bar{\xi}, \bar{\eta} \in \mathcal{F}_o} h(\bar{\eta}, \bar{\xi}).$$

(N.B. one cannot simply take for $\text{Fl } \mathcal{F}$ the *union* of the sets $h(\bar{\eta}, \bar{\xi})$, for the latter are not necessarily disjoint). Moreover:

- 3) The maps $p_o: \text{Ob } \mathcal{F} \rightarrow \text{Ob } \mathcal{E}$ and $p_1 = (p_{\bar{\eta}, \bar{\xi}}): \text{Fl } \mathcal{F} \rightarrow \text{Fl } \mathcal{E}$ define a *functor* $p: \mathcal{F} \rightarrow \mathcal{E}$. In this way, $\mathcal{F}$ becomes a category over $\mathcal{E}$, and moreover the obvious map $h_f(\bar{\eta}, \bar{\xi}) \rightarrow \text{Hom}(\bar{\eta}, \bar{\xi})$ induces a *bijection*

$$h_f(\bar{\eta}, \bar{\xi}) \xrightarrow{\sim} \text{Hom}_f(\bar{\eta}, \bar{\xi}).$$

- 4) The obvious maps

$$\text{Ob } \mathcal{F}(S) \longrightarrow \mathcal{F}_o = \text{Ob } \mathcal{F}, \quad \text{Fl } \mathcal{F}(S) \longrightarrow \text{Fl } \mathcal{F},$$

where the second is defined by the obvious maps

$$\text{Hom}_{\mathcal{F}(S)}(\xi, \xi') = h_{\text{id}_S}(\bar{\xi}, \bar{\xi}') \longrightarrow \text{Hom}(\bar{\xi}, \bar{\xi}')$$

define an *isomorphism*

$$i_S: \mathcal{F}(S) \xrightarrow{\sim} \mathcal{F}_S.$$

- 5) For every object $\bar{\xi} = (S, \xi)$ of $\mathcal{F}$, and every morphism $f: T \rightarrow S$ of $\mathcal{E}$, consider the element $\bar{\eta} = (T, \eta)$ of $\mathcal{F}_T$, with $\eta = f^*(\xi)$, and the element $\alpha_f(\xi)$ of $\text{Hom}(\bar{\eta}, \bar{\xi})$, image of $\text{id}_{f^*(\xi)}$ by the morphism $\text{Hom}_{\mathcal{F}(T)}(f^*(\xi), f^*(\xi)) = h_f(\bar{\eta}, \bar{\xi}) \rightarrow \text{Hom}_f(\bar{\eta}, \bar{\xi})$. *This element is cartesian, and it is the identity in $\bar{\xi}$ if $f = \text{id}_S$* , in other words, the set of the $\alpha_f(\xi)$ defines a *normalized cleavage of $\mathcal{F}$ over $\mathcal{E}$* . Moreover, by construction, one has commutativity in the diagram of functors

$$\begin{array}{ccc} \mathcal{F}(S) & \xrightarrow{f^*} & \mathcal{F}(T) \\ i_S \downarrow & & \downarrow i_T \\ \mathcal{F}_S & \xrightarrow{f_{\mathcal{F}}^*} & \mathcal{F}_T \end{array}$$

where $f_{\mathcal{F}}^*$ is the inverse image functor by f , relative to the cleavage under consideration on $\mathcal{F}$. Finally:

- 6) the homomorphisms $c_{f,g}$ given with the pseudofunctor are transformed, by the isomorphisms i_S , into the functorial homomorphisms $c_{f,g}$ associated to the cleavage of $\mathcal{F}$.

We confine ourselves to giving the verification of 1) (which is, if possible, less trivial than the others). It suffices to prove the associativity of the composition between the objects of sets of the form $h_f(\bar{\eta}, \bar{\xi})$. Consider therefore in $\mathcal{E}$ morphisms

$$S \xleftarrow{f} T \xleftarrow{g} U \xleftarrow{h} V$$

and objects

$$\xi, \eta, \zeta, \tau$$

in $\mathcal{F}(S), \mathcal{F}(T), \mathcal{F}(U), \mathcal{F}(V)$, finally elements

$$\begin{aligned} u \in h_f(\bar{\eta}, \bar{\xi}) &= \text{Hom}_{\mathcal{F}(T)}(\eta, f^*(\xi)) \\ v \in h_g(\bar{\zeta}, \bar{\eta}) &= \text{Hom}_{\mathcal{F}(U)}(\zeta, g^*(\eta)) \\ w \in h_h(\bar{\tau}, \bar{\zeta}) &= \text{Hom}_{\mathcal{F}(V)}(\tau, h^*(\zeta)). \end{aligned}$$

One wants to prove the formula

$$(u \circ v) \circ w = u \circ (v \circ w),$$

which is an equality in $\text{Hom}_{\mathcal{F}(V)}(\tau, (fgh)^*(\xi))$. By virtue of the definitions the two sides of this equality are obtained by composition along the upper and lower contours of the diagram below:

$$\begin{array}{ccccccc}
 & & & & h^*(u \circ v) & & \\
 & & \swarrow & & \searrow & & \\
 \tau & \xrightarrow{w} & h^*(\zeta) & \xrightarrow{h^*(v)} & h^*g^*(\eta) & \xrightarrow{h^*g^*(u)} & h^*g^*f^*(\xi) \xrightarrow{h^*(c_{f,g}(\xi))} h^*(fg)^*(\xi) \\
 & \searrow & \downarrow c_{g,h}(\eta) & & \downarrow c_{g,h}(f^*(\xi)) & & \downarrow c_{fg,h}(\xi) \\
 & & (gh)^*(\eta) & \xrightarrow{(gh)^*(u)} & (gh)^*f^*(\xi) & \xrightarrow{c_{f,gh}(\xi)} & (fgh)^*(\xi)
 \end{array}$$

Now the middle square is commutative because $c_{g,h}$ is a functorial homomorphism, and the right-hand square is commutative by virtue of the condition B) for a pseudofunctor. Whence the announced result.

Of course, it remains to specify, when the pseudofunctor under consideration already comes from a normalized cloven category $\mathcal{F}'$ over $\mathcal{E}$, how one obtains a natural isomorphism between $\mathcal{F}'$ and $\mathcal{F}$. We leave the details to the reader.

We also leave to the reader to interpret, in terms of pseudofunctors, the notion of inverse image of a cloven category $\mathcal{F}$ over $\mathcal{E}$ by a change of base functor $\mathcal{E}' \rightarrow \mathcal{E}$.

9. EXAMPLE: CLOVEN CATEGORY DEFINED BY A FUNCTOR $\mathcal{E}^\circ \rightarrow \mathbf{Cat}$; SPLIT CATEGORIES OVER $\mathcal{E}$

Suppose one has a functor

$$\varphi: \mathcal{E}^\circ \longrightarrow \mathbf{Cat},$$

it then defines a pseudofunctor by setting

$$\mathcal{F}(S) = \varphi(S), \quad f^* = \varphi(f), \quad c_{f,g} = \text{id}_{(fg)^*}$$

Thus the construction of the preceding no. gives us a category $\mathcal{F}$ cloven over $\mathcal{E}$, said to be associated to the functor φ . For a cloven category over $\mathcal{E}$ to be isomorphic to a cloven category defined by a functor $\varphi: \mathcal{E}^\circ \rightarrow \mathbf{Cat}$, it is necessary and sufficient, manifestly, that it satisfy the conditions:

$$(fg)^* = g^*f^*, \quad c_{f,g} = \text{id}_{(fg)^*}.$$

In terms of the set K of transport morphisms, this also simply means that *the composite of two transport morphisms is a transport morphism*. A cleavage of a category $\mathcal{F}$ over $\mathcal{E}$ satisfying the preceding condition is called a *splitting* of $\mathcal{F}$ over $\mathcal{E}$, and a category $\mathcal{F}$ over $\mathcal{E}$ equipped with a splitting is called a *split category over $\mathcal{E}$* . It is thus a particular case of the notion of cloven category. The category of split categories over $\mathcal{E}$ is thus equivalent to $\mathbf{Hom}(\mathcal{E}^\circ, \mathbf{Cat})$. Note that a split category over $\mathcal{E}$ is a fortiori a cloven category over $\mathcal{E}$.

If $\mathcal{F}$ is a fibered category over $\mathcal{E}$, there does not always exist a splitting on $\mathcal{F}$. Suppose for example that $\text{Ob } \mathcal{E}$ and $\text{Ob } \mathcal{F}$ are reduced to a single element, and that the set of endomorphisms of the said element is a group E resp. F , so that the projection functor p is given by a homomorphism of groups $p: F \rightarrow E$, surjective since p is fibrant. One then verifies at once that the set of cleavages of $\mathcal{F}$ over $\mathcal{E}$ is in bijective correspondence with the set of maps $s: E \rightarrow F$ such that $ps = \text{id}_E$ (i.e. the set of “systems of representatives” for the classes mod the subgroup G kernel of the surjective homomorphism $p: F \rightarrow E$). A cleavage is a splitting if and only if s is a homomorphism of groups. To say that there exists a splitting therefore means that the group extension F of E by G is trivial, which is expressed, when G is commutative, by the vanishing of a certain cohomology class in $H^2(E, G)$ (where G is regarded as a group on which E acts).

Suppose however that $\mathcal{F}$ is a fibered category over $\mathcal{E}$ such that the $\mathcal{F}_S$ are *rigid* categories, i.e. the group of automorphisms of every object of $\mathcal{F}_S$ is reduced to the identity. It is then

easy to prove that $\mathcal{F}$ admits a splitting over $\mathcal{E}$. Indeed, one first observes that the question of existence of a splitting is not modified if one replaces $\mathcal{F}$ by an $\mathcal{E}$ -equivalent category, which in the present instance brings us back to the case where the $\mathcal{F}_S$ are rigid *and reduced* categories (i.e. two isomorphic objects in $\mathcal{F}_S$ are identical). But if G is a rigid and reduced category, every isomorphism of two functors $H \rightarrow G$ (where H is an arbitrary category) is an identity. It follows that if $\mathcal{F}$ is a fibered category over $\mathcal{E}$, such that the fiber-categories are rigid and reduced, then there exists a *unique* cleavage of $\mathcal{F}$ over $\mathcal{E}$, which is necessarily a splitting. Thus $\mathcal{F}$ is isomorphic to the category defined by a functor $\varphi: \mathcal{E}^\circ \rightarrow \mathbf{Cat}$, such that the $\varphi(S)$ are rigid and discrete categories, and the functor φ is defined up to isomorphism.

10. COFIBERED CATEGORIES, BIFIBERED CATEGORIES

Let us consider a category $\mathcal{F}$ over $\mathcal{E}$, with the projection functor

$$p: \mathcal{F} \longrightarrow \mathcal{E},$$

it defines a category $\mathcal{F}^\circ$ over $\mathcal{E}^\circ$, by the projection functor

$$p^\circ: \mathcal{F}^\circ \longrightarrow \mathcal{E}^\circ.$$

A morphism $\alpha: \eta \rightarrow \xi$ in $\mathcal{F}$ is said to be *cocartesian* if it is a cartesian morphism for $\mathcal{F}^\circ$ over $\mathcal{E}^\circ$. Making this explicit, one sees that this means that for every object ξ' of $\mathcal{F}_S$, the map $u \mapsto u \circ \alpha$

$$\mathrm{Hom}_S(\xi, \xi') \longrightarrow \mathrm{Hom}_f(\eta, \xi')$$

is bijective. One then also says that (ξ, α) is a *direct image* of η by f , in the category $\mathcal{F}$ over $\mathcal{E}$. If it exists for every η in $\mathcal{F}_T$, one says that the direct image functor by f exists, and one denotes this functor $f_*^{\mathcal{F}}$ or f_* , once it has been chosen. It is thus defined by an isomorphism of bifunctors on $\mathcal{F}_T^\circ \times \mathcal{F}_S$:

$$\mathrm{Hom}_S(f_*(\eta), \xi) \xrightarrow{\sim} \mathrm{Hom}_f(\eta, \xi).$$

Thus if f_* exists, for f^* to exist, it is necessary and sufficient that f_* admit an adjoint functor, i.e. that there exist a functor $f^*: \mathcal{F}_S \rightarrow \mathcal{F}_T$ and an isomorphism of bifunctors

$$\mathrm{Hom}_S(f_*(\eta), \xi) \xrightarrow{\sim} \mathrm{Hom}_T(\eta, f^*(\xi)).$$

Let $g: U \rightarrow T$ be another morphism in $\mathcal{E}$, and suppose that the inverse images and direct images by f, g and fg exist. Let us consider then the functorial homomorphisms

$$\begin{aligned} c_{f,g}^f: f_* g_* &\longleftarrow (fg)_* \\ c_{f,g}: g^* f^* &\longrightarrow (fg)^*. \end{aligned}$$

One observes that if one regards $f_* g_*$ and $g^* f^*$ as a pair of adjoint functors, as well as $(fg)_*$ and $(fg)^*$, the two preceding homomorphisms are adjoint to one another. Thus one is an isomorphism if and only if the other is. In particular:

Proposition 10.1. *Suppose that the category $\mathcal{F}$ over $\mathcal{E}$ is prefibered and coprefibered. For it to be fibered, it is necessary and sufficient that it be cofibered.*

Of course, one says that $\mathcal{F}$ is coprefibered resp. cofibered over $\mathcal{E}$, if $\mathcal{F}^\circ$ is prefibered resp. fibered over $\mathcal{E}$. We shall say that $\mathcal{F}$ is bifibered over $\mathcal{E}$, if it is at once fibered and cofibered over $\mathcal{E}$.

11. VARIOUS EXAMPLES

- a) **Categories of arrows of $\mathcal{E}$.** Let $\mathcal{E}$ be a category. Let us denote by Δ^1 the category associated to the totally ordered set with two elements $[0, 1]$; it thus has two objects 0 and 1, and besides the two identity morphisms an arrow $(0, 1)$ with source 0 and target 1. Let

$$\mathbf{Fl}(\mathcal{E}) = \mathbf{Hom}(\Delta^1, \mathcal{E})$$

one calls it the *category of arrows of $\mathcal{E}$* . The object 1 of Δ^1 defines a canonical functor, called the *target functor*

$$\mathbf{Fl}(\mathcal{E}) \longrightarrow \mathcal{E}$$

(the functor defined by the object 0 of Δ^1 is called the *source functor*). For every object S of $\mathcal{E}$, the fiber-category $\mathbf{Fl}(\mathcal{E})_S$ is canonically isomorphic to the category $\mathcal{E}_{/S}$ of objects of $\mathcal{E}$ over S .

Let us consider a morphism $f: T \rightarrow S$ in $\mathcal{E}$, then there corresponds to it a canonical functor

$$f_*: \mathcal{E}_{/T} = \mathcal{F}_T \longrightarrow \mathcal{E}_{/S} = \mathcal{F}_S$$

and a functorial isomorphism

$$\mathrm{Hom}_S(f_*(\eta), \xi) \xrightarrow{\sim} \mathrm{Hom}_f(\eta, \xi)$$

which thus makes of f_* a direct image functor for f in $\mathcal{F}$. One has moreover here

$$(\mathrm{id}_S)_* = \mathrm{id}_{\mathcal{F}_S}, \quad (fg)_* = f_*g_*, \quad c^{f,g} = \mathrm{id}_{(fg)},$$

i.e. $\mathcal{F}$ is equipped with a co-splitting over $\mathcal{E}$. A fortiori, $\mathcal{F}$ is cofibered over $\mathcal{E}$. Let us now note that the set of morphisms in $\mathcal{F}$ is in bijective correspondence with the set of commutative square diagrams in $\mathcal{E}$.

$$\begin{array}{ccc} X & \xleftarrow{f'} & Y \\ u \downarrow & & \downarrow v \\ S & \xleftarrow{f} & T \end{array}$$

By definition, the morphism in question is cartesian if the square is cartesian in $\mathcal{E}$, i.e. if it makes of Y a fibered product of X and T over S . The inverse image functor f^* thus exists if and only if for every object X over S , the fibered product $X \times_S T$ exists. It follows from 10.1 that if the product of two objects over a third always exists in $\mathcal{E}$, i.e. if $\mathcal{F}$ is prefibered over $\mathcal{E}$, then $\mathcal{F}$ is even fibered over $\mathcal{E}$.

b) Category of presheaves or sheaves on variable spaces

Let $\mathcal{E} = \mathbf{Top}$ be the category of topological spaces. If T is a topological space, we shall denote by $\mathcal{U}(T)$ the category of open sets of T , where the morphisms are the inclusion maps. If $\mathcal{C}$ is a category, a functor $\mathcal{U}(T)^\circ \rightarrow \mathcal{C}$ is called a *presheaf* on T with values in $\mathcal{C}$, and a *sheaf* if it satisfies a left exactness condition which we do not repeat here. The *category $\mathcal{P}(T)$ of presheaves on T with values in $\mathcal{C}$* is by definition the category $\mathbf{Hom}(\mathcal{U}(T)^\circ, \mathcal{C})$, and the category $\mathcal{F}(T)$ of sheaves on T with values in $\mathcal{C}$ is the full subcategory whose objects are the objects of $\mathbf{Hom}(\mathcal{U}(T)^\circ, \mathcal{C})$ which are sheaves. If $f: T \rightarrow S$ is a morphism in $\mathcal{E}$, i.e. a continuous map of topological spaces, there corresponds to it by the increasing map $U \mapsto f^{-1}(U)$ a functor $\mathcal{U}(S) \rightarrow \mathcal{U}(T)$, whence a functor

$$f_*: \mathbf{Hom}(\mathcal{U}(T)^\circ, \mathcal{C}) \longrightarrow \mathbf{Hom}(\mathcal{U}(S)^\circ, \mathcal{C})$$

called the *direct image functor of presheaves by f* . One sees immediately that the direct image of a sheaf is a sheaf, thus the functor $f_*: \mathcal{P}(T) \rightarrow \mathcal{P}(S)$ induces a functor, likewise denoted $f_*: \mathcal{F}(T) \rightarrow \mathcal{F}(S)$. One moreover verifies trivially (by the associativity of the composition of functors) that one has, for a second continuous map $g: U \rightarrow T$, the identity

$$(gf)_* = g_*f_*, \quad \text{likewise } (\mathrm{id}_S)_* = \mathrm{id}_{\mathcal{P}(S)}.$$

In this way, one has obtained a functor

$$S \longmapsto \mathcal{P}(S)$$

resp.

$$S \longmapsto \mathcal{F}(S)$$

from $\mathcal{E}$ to **Cat**. In fact, we are interested in the corresponding functor

$$S \longmapsto \mathcal{P}(S)^\circ, \quad \text{resp. } S \longmapsto \mathcal{F}(S)^\circ.$$

It defines a cofibered category, and even a co-split one, over the category of topological spaces, which one calls the *cofibered category of presheaves* (resp. *sheaves*) *with values in $\mathcal{C}$* (understood: on variable spaces). Making explicit the construction of no. 8, one sees that a morphism from a presheaf B on T to a presheaf A on S is a pair (f, u) consisting of a continuous map from T to S , and of a morphism $u: A \rightarrow f_*(B)$ in the category $\mathcal{P}(S)$. This description is equally valid for morphisms of sheaves, $\mathcal{F}$ being a full subcategory of $\mathcal{P}$.

In the most important cases, the category $\mathcal{P}$ and the category $\mathcal{F}$ over $\mathcal{E}$ are also fibered categories, i.e. for every continuous map, the direct image functors $\mathcal{P}(T) \rightarrow \mathcal{P}(S)$ and $\mathcal{F}(T) \rightarrow \mathcal{F}(S)$ have an adjoint functor, which is then denoted f^* and called the inverse image functor of presheaves resp. the inverse image functor of sheaves, by the continuous map f . This functor exists for example if $\mathcal{C} = \mathbf{Ens}$. One can show that the functor $f^*: \mathcal{P}(S) \rightarrow \mathcal{P}(T)$ exists whenever in $\mathcal{C}$ the inductive limits (relative to diagrams in the Universe under consideration) exist. The question is less easy for $\mathcal{F}$; indeed, one notes that (even in the case $\mathcal{C} = \mathbf{Ens}$) the inverse image of a presheaf which is a sheaf is in general not a sheaf, in other words the inverse image functor of sheaves is not isomorphic to the functor induced by the inverse image functor of presheaves (despite the common notation f^*). Thus, $\mathcal{F}$ is a cofibered subcategory of $\mathcal{P}$, but not a fibered subcategory, i.e. *the inclusion functor $\mathcal{F} \rightarrow \mathcal{P}$ is not fibrant*.

The cofibered category $\mathcal{P}$ can be deduced from a more general cofibered (or rather fibered) category, obtained as follows. For every category $\mathcal{U}$ (in the fixed Universe), one sets

$$\mathcal{P}(\mathcal{U}) = \mathbf{Hom}(\mathcal{U}, \mathcal{C})$$

and one notes that $\mathcal{U} \mapsto \mathcal{P}(\mathcal{U})$ is in a natural way a contravariant functor in $\mathcal{U}$, from the category **Cat** to **Cat**. It thus defines a split category over $\mathcal{E} = \mathbf{Cat}$, which we shall denote $\mathbf{Cat}_{//\mathcal{C}}$. The objects of this category are the pairs $(\mathcal{U}, p)$ of a category $\mathcal{U}$ and a functor $p: \mathcal{U} \rightarrow \mathcal{C}$, and a morphism from $(\mathcal{U}, p)$ to $(\mathcal{V}, q)$ is essentially a pair (f, u) , where f is a functor $\mathcal{U} \rightarrow \mathcal{V}$ and u a homomorphism of functors $u: p \rightarrow qf$. We leave to the reader the task of making explicit the composition of morphisms in $\mathbf{Cat}_{//\mathcal{C}}$. The projection functor

$$\mathcal{F} = \mathbf{Cat}_{//\mathcal{C}} \longrightarrow \mathcal{E} = \mathbf{Cat}$$

associates to the pair $(\mathcal{U}, p)$ the object $\mathcal{U}$; the fiber-category at $\mathcal{U}$ is the category $\mathbf{Hom}(\mathcal{U}, \mathcal{C})$ (up to isomorphism). When inductive limits exist in $\mathcal{C}$, one shows easily that the fibered category $\mathbf{Cat}_{//\mathcal{C}}$ over **Cat** is likewise cofibered over **Cat**, i.e. one can define the notion of *direct image of a functor $p: \mathcal{U} \rightarrow \mathcal{C}$* by a functor $f: \mathcal{U} \rightarrow \mathcal{V}$. The category of presheaves is deduced from the preceding fibered category by the change of base

$$\mathbf{Top}^\circ \longrightarrow \mathbf{Cat}$$

(functor $S \mapsto \mathcal{U}(S)$ defined above), which gives a fibered category over $\mathbf{Top}^\circ$, and passing to the opposite category, one obtains the cofibered category $\mathcal{P}$ of presheaves over **Top**. The notion of inverse image of a functor corresponds to that of direct image of a presheaf, the notion of direct image of a functor to that of inverse image of a presheaf.

c) Objects with operators over an object with operators

Let $\mathcal{F}$ be a category over $\mathcal{E}$, and let S be an object of $\mathcal{E}$ on which a group G operates, on the left to fix ideas. This object with operators can be interpreted as corresponding to a functor $\lambda: \mathcal{E}' \rightarrow \mathcal{E}$ from the category (with a single object, having

G as group of endomorphisms) $\mathcal{E}'$ defined by G , into the category $\mathcal{E}$, and thus there is defined by change of base a category $\mathcal{F}'$ over $\mathcal{E}'$, which is fibered resp. cofibered when $\mathcal{F}$ is so over $\mathcal{E}$. A section of $\mathcal{E}'$ over $\mathcal{F}'$ (necessarily cartesian, since $\mathcal{E}'$ is a groupoid, and every isomorphism in $\mathcal{F}'$ is cartesian by virtue of 6.12), can also be interpreted as an $\mathcal{E}$ -functor $\mathcal{E}' \rightarrow \mathcal{F}$ over λ , or also as an object with operators ξ in $\mathcal{F}$ “over” the object with operators S .

d) **Pairs of quasi-inverse adjoint functors; autodualities**

When the base-category $\mathcal{E}$ is reduced to two objects a, b and, besides the identity arrows, to two isomorphisms $f: a \rightarrow b$ and $g: b \rightarrow a$ inverse to one another (i.e. $\mathcal{E}$ is a connected rigid groupoid with two objects), a normalized cloven category over $\mathcal{E}$ is essentially the same thing as the system formed by two categories $\mathcal{F}_a$ and $\mathcal{F}_b$ and a pair of adjoint functors $G: \mathcal{F}_a \rightarrow \mathcal{F}_b$ and $F: \mathcal{F}_b \rightarrow \mathcal{F}_a$, which are equivalences of categories (hence quasi-inverse to one another). One will take for $\mathcal{F}_a$ and $\mathcal{F}_b$ the fiber-categories of $\mathcal{F}$, for F and G the functors f^* and g^* , and the two isomorphisms

$$u: FG \xrightarrow{\sim} \text{id}_{\mathcal{F}_a} \quad v: GF \xrightarrow{\sim} \text{id}_{\mathcal{F}_b}$$

are $c_{g,f}$ and $c_{f,g}$. The two usual compatibility conditions between u and v are none other than the condition 7.4 B) for the composites $f g f$ and $g f g$. It is easy to show that these conditions suffice to imply that one indeed has a pseudofunctor $\mathcal{E}^\circ \rightarrow \mathbf{Cat}$.

An interesting case is that where one has

$$\mathcal{F}_b = \mathcal{F}_a^\circ, \quad G = F^\circ, \quad v = u^\circ.$$

One calls *autoduality* in a category $\mathcal{C}$, the data of a functor $D: \mathcal{C} \rightarrow \mathcal{C}^\circ$, and of an isomorphism $u: DD^\circ \xrightarrow{\sim} \text{id}_{\mathcal{C}}$, such that u and the isomorphism $u^\circ: D^\circ D \xrightarrow{\sim} \text{id}_{\mathcal{C}^\circ}$ make of (D, D°) a pair of adjoint functors, (necessarily quasi-inverse to one another). This condition is written:

$$D(u(x)) = u(D(x)) \quad \text{for every } x \in \text{Ob}(\mathcal{C}).$$

- e) **Categories over a discrete category $\mathcal{E}$.** One says that $\mathcal{E}$ is a *discrete category* if every arrow therein is an identity arrow, so that $\mathcal{E}$ is defined up to unique isomorphism by the knowledge of the set $I = \text{Ob}(\mathcal{E})$. The data of a category $\mathcal{F}$ over $\mathcal{E}$ is thus equivalent (up to unique isomorphism) to the data of a family of categories $\mathcal{F}_i$ ($i \in I$), the fiber-categories. Every category $\mathcal{F}$ over $\mathcal{E}$ is fibered, every $\mathcal{E}$ -functor $\mathcal{F} \rightarrow \mathcal{G}$ is cartesian, one has a canonical isomorphism

$$\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{G}) \xrightarrow{\sim} \prod_i \mathbf{Hom}(\mathcal{F}_i, \mathcal{G}_i).$$

In particular, one obtains

$$\Gamma(\mathcal{F}/\mathcal{E}) = \varprojlim \mathcal{F}/\mathcal{E} \xrightarrow{\sim} \prod_i \mathcal{E}_i.$$

- f) **Suppose that $\mathcal{E}$ has exactly two objects S and T ,** and besides the **identity morphisms, a morphism $f: T \rightarrow S$.** Then a category $\mathcal{F}$ over $\mathcal{E}$ is defined, up to unique $\mathcal{E}$ -isomorphism, by the data of two categories $\mathcal{F}_S$ and $\mathcal{F}_T$ and of a bifunctor $H(\eta, \xi)$ on $\mathcal{F}_T^\circ \times \mathcal{F}_S$, with values in **Ens**. Indeed, if $\mathcal{F}$ is a category over $\mathcal{E}$, one associates to it the two fiber-categories $\mathcal{F}_S$ and $\mathcal{F}_T$ and the bifunctor $H(\eta, \xi) = \text{Hom}_{\mathcal{F}}(\eta, \xi)$. One leaves to the reader the task of making explicit the construction in the opposite direction. For the category under consideration to be fibered (or prefibered, which amounts to the same) it is necessary and sufficient that the functor H be representable with respect to the argument ξ . For it to be cofibered, it is necessary and sufficient that H be representable with respect to the argument η .
- g) Let $\mathcal{F} = \mathcal{C} \times \mathcal{E}$, considered as a category over $\mathcal{E}$ by means of pr_2 . Then $\mathcal{F}$ is fibered and cofibered over $\mathcal{E}$, and is even equipped with a canonical splitting and co-splitting,

corresponding to the constant functor on $\mathcal{E}$, resp. on $\mathcal{E}^\circ$, with values in $\mathbf{Cat}$, of value $\mathcal{C}$. One has

$$\Gamma(\mathcal{F}/\mathcal{E}) \simeq \mathbf{Hom}(\mathcal{E}, \mathcal{C})$$

and $\varprojlim \mathcal{F}/\mathcal{E}$ corresponds to the full subcategory formed of the functors $F: \mathcal{E} \rightarrow \mathcal{C}$ transforming arbitrary morphisms into isomorphisms.

12. FUNCTORS ON A CLOVEN CATEGORY

Let $\mathcal{F}$ be a normalized cloven category over $\mathcal{E}$. For every object S of $\mathcal{E}$ one denotes by

$$i_S: \mathcal{F}_S \longrightarrow \mathcal{F}$$

the inclusion functor. One thus has a functorial homomorphism, for every morphism $f: T \rightarrow S$ in $\mathcal{E}$:

$$\alpha_f: i_T f^* \longrightarrow i_S,$$

where f^* is the change of base functor $\mathcal{F}_S \rightarrow \mathcal{F}_T$ for f defined by the cleavage. Let now

$$F: \mathcal{F} \longrightarrow \mathcal{C}$$

be a functor from $\mathcal{F}$ to a category $\mathcal{C}$; set, for every $S \in \text{Ob}(\mathcal{E})$,

$$F_S = F \circ i_S: \mathcal{F}_S \longrightarrow \mathcal{C}$$

and for every $f: T \rightarrow S$ in $\mathcal{E}$,

$$\varphi_f = F * \alpha_f: F_T f^* \longrightarrow F_S$$

One has thus, to every functor $F: \mathcal{F} \rightarrow \mathcal{C}$, associated a family (F_S) of functors $\mathcal{F}_S \rightarrow \mathcal{C}$, and a family (φ_f) of homomorphisms of functors $F_T f^* \rightarrow F_S$. These families satisfy the following conditions:

- a) $\varphi_{\text{id}_S} = \text{id}_{F_S}$.
- b) For two morphisms $f: T \rightarrow S$ and $g: U \rightarrow T$ in $\mathcal{E}$, one has commutativity in the square of functorial homomorphisms:

$$\begin{array}{ccc} F_U g^* f^* & \xrightarrow{F_U * c_{f,g}} & F_U (fg)^* \\ \varphi_g * f^* \downarrow & & \downarrow \varphi_{fg} \\ F_T f^* & \xrightarrow{\varphi_f} & F_S. \end{array}$$

The first relation is trivial, and the second relation is obtained by applying the functor F to the commutative diagram

$$\begin{array}{ccc} g^* f^*(\xi) & \xrightarrow{c_{f,g}(\xi)} & (fg)^*(\xi) \\ \alpha_g(f^*(\xi)) \downarrow & & \downarrow \alpha_{fg}(\xi) \\ f^*(\xi) & \xrightarrow{\alpha_f(\xi)} & \xi \end{array}$$

for a variable object ξ in $\mathcal{F}_S$.

If G is a second functor $\mathcal{F} \rightarrow \mathcal{C}$, giving rise to functors $G_S: \mathcal{F}_S \rightarrow \mathcal{C}$ and functorial homomorphisms $\psi_f: G_T f \rightarrow G_S$, and if $u: F \rightarrow G$ is a functorial homomorphism, then there correspond to it functorial homomorphisms $u * i_S$:

$$u_S: F_S \longrightarrow G_S$$

and one observes immediately that for every morphism $f: T \rightarrow S$ in $\mathcal{E}$, one has commutativity in the squares

$$\text{c) } \begin{array}{ccc} F_T f^* & \xrightarrow{\phi_f} & F_S \\ u_T^* f^* \downarrow & & \downarrow u_S \\ G_T f^* & \xrightarrow{\psi_f} & G_S \end{array}$$

Proposition 12.1. *Let $\mathcal{H}(\mathcal{F}, \mathcal{C})$ be the category whose objects are the pairs of families (F_S) ($S \in \text{Ob}(\mathcal{F})$) of functors $\mathcal{F}_S \rightarrow \mathcal{C}$, and of families (φ_f) ($f \in \text{Fl}(\mathcal{F})$) of functorial homomorphisms $F_T f^* \rightarrow F_S$, satisfying the conditions a) and b), and where the morphisms are the families (u_S) ($S \in \text{Ob}(\mathcal{F})$) of homomorphisms $F_S \rightarrow G_S$, verifying the commutativity condition c) written above, (the composition of morphisms being effected by the composition of the homomorphisms of functors $\mathcal{F}_S \rightarrow \mathcal{C}$). Then the two laws made explicit above define an isomorphism K of the category $\mathbf{Hom}(\mathcal{F}, \mathcal{C})$ with the category $\mathcal{H}(\mathcal{F}, \mathcal{C})$.*

It is trivial that one has indeed here a *functor* from the first category to the second. This functor is fully faithful, since for given F, G , $\text{Hom}(F, G) \rightarrow \text{Hom}(K(F), K(G))$ is trivially injective; to show that it is surjective, it suffices to note that the commutativity condition c) expresses the functoriality of the maps $u(\xi) = u_S(\xi): F(S) = F_S(\xi) \rightarrow G(\xi) = G_S(\xi)$ for the homomorphisms of the form $\alpha_f(\xi)$ in $\mathcal{F}$, on the other hand one has functoriality on each fiber-category i.e. for the morphisms in $\mathcal{F}$ which are T -morphisms ($T \in \text{Ob}(\mathcal{E})$), whence functoriality for every morphism in $\mathcal{F}$, since an f -morphism (where $f: T \rightarrow S$ is a morphism in $\mathcal{E}$ is in a unique way) a composite of a morphism $\alpha_f(\xi)$ and a T -morphism. It remains therefore to prove that the functor K is bijective on objects. The preceding argument already shows that K is injective on objects; it remains to prove that it is surjective, i.e. that if one starts from a system $(F_S), (\varphi_f)$, satisfying a) and b); and if one defines a map $\text{Ob } \mathcal{F} \rightarrow \text{Ob } \mathcal{C}$ by

$$F(\xi) = F_S(\xi) \quad \text{for } \xi \in \text{Ob } \mathcal{F}_S \subset \text{Ob } \mathcal{F}$$

and a map $\text{Fl}(\mathcal{F}) \rightarrow \text{Fl}(\mathcal{C})$ by

$$F(\alpha_f(\xi)u') = \varphi_f(\xi) F_T(u')$$

for every morphism $f: T \rightarrow S$ in $\mathcal{E}$, every object ξ of $\mathcal{F}_S$ and every T -morphism u' with target $f^*(\xi)$, then one obtains a *functor* F from $\mathcal{F}$ to $\mathcal{C}$. Indeed, the relation $F(\text{id}_\xi) = \text{id}_{F(\xi)}$ is trivial, it remains to prove the multiplicativity $F(uv) = F(u)F(v)$ when one has an f -morphism $u: \eta \rightarrow \xi$ and a g -morphism $v: \zeta \rightarrow \eta$, with $f: T \rightarrow S$ and $g: U \rightarrow T$ morphisms of $\mathcal{E}$. Setting $w = uv$, one will have

$$u = \alpha_f(\xi)u', \quad v = \alpha_g(\eta)v', \quad w = \alpha_{fg}(\xi)w'$$

with

$$w' = c_{f,g}(\xi)g^*(u')v' \quad (\text{cf. no. 8}).$$

With these notations, one must prove the commutativity of the outer contour of the diagram below:

$$\begin{array}{ccccccc} & & & & F(w') & & \\ & & & & \uparrow & & \\ & & & & \text{---} & & \\ F_U(\zeta) & \xrightarrow{F_u(v')} & F_U g^*(\eta) & \xrightarrow{F_u g^*(u')} & F_U g^* f^*(\xi) & \xrightarrow{F_U(c_{f,g}(\xi))} & F_U(fg)^*(\xi) \\ & \searrow F(v) & \downarrow \varphi_g(\eta) & & \downarrow \varphi_g(f^*(\xi)) & & \downarrow \varphi_{fg}(\xi) \\ & & F_T(\eta) & \xrightarrow{F_T(u')} & F_T f^*(\xi) & \xrightarrow{\varphi_f(\xi)} & F_S(\xi) \\ & & & & \text{---} & & \\ & & & & F(u) & & \end{array}$$

Now the left triangle is commutative by definition of $F(v)$, the middle square is commutative since it is deduced from the homomorphism $u': \xi \rightarrow f(\eta)$ by the functorial homomorphism α_g ,

finally the right-hand square is commutative by virtue of condition b). The desired conclusion follows.

Suppose now that $\mathcal{C}$ is likewise a normalized cloven category over $\mathcal{E}$, which we shall henceforth call $\mathcal{G}$, and that we are interested in the $\mathcal{E}$ -functors from $\mathcal{F}$ to $\mathcal{G}$. If F is such a functor, it induces functors

$$F_S: \mathcal{F}_S \longrightarrow \mathcal{G}_S$$

for the fiber-categories. On the other hand, for every morphism $f: T \rightarrow S$ in $\mathcal{E}$, and every object ξ in $\mathcal{F}_S$, the f -morphism $F(\alpha_f(\xi))$ factors in a unique way through a T -morphism

$$\varphi_f(\xi): F_T(f_{\mathcal{F}}^*(\xi)) \longrightarrow f_{\mathcal{G}}^*(F_S(\xi))$$

(where the $\mathcal{F}$ or the $\mathcal{G}$ in the index indicates the cloven category for which one takes the inverse image functor), whence a functorial homomorphism of functors from $\mathcal{F}_S$ to $\mathcal{G}_T$:

$$\varphi_f: F_T f_{\mathcal{F}}^* \longrightarrow f_{\mathcal{G}}^* F_S.$$

The two systems (F_S) and (φ_f) satisfy the following conditions:

a') $\varphi_{\text{id}_S} = \text{id}_{F_S}$.

b') For two morphisms $f: T \rightarrow S$ and $g: U \rightarrow T$ in $\mathcal{E}$, one has commutativity in the following diagram of functorial homomorphisms:

$$\begin{array}{ccc} F_U g_F^* f_F^* & \xrightarrow{F_U * c_{f,g}^{\mathcal{F}}} & F_U (fg)_F^* \\ \varphi_g^* f_{\mathcal{F}}^* \downarrow & & \downarrow \varphi_{fg} \\ g_{\mathcal{G}}^* F_T f_F^* & & \\ g_{\mathcal{G}}^* \varphi_f \downarrow & & \\ g_{\mathcal{G}}^* f_{\mathcal{G}}^* F_S & \xrightarrow{c_{f,g}^{\mathcal{G}} * F_S} & (fg)_{\mathcal{G}}^* F_S. \end{array}$$

We leave the verification to the reader, as well as the statement and the proof of the analogue of proposition 12.1, implying that one thus obtains a bijective correspondence between the set of $\mathcal{E}$ -functors from $\mathcal{F}$ to $\mathcal{G}$, and the set of systems $(F_S), (\varphi_f)$ satisfying the conditions a') and b') above. Of course, in this correspondence, the cartesian functors are characterized by the property that the homomorphisms φ_f are isomorphisms.

Remark. Of course, it is most often advantageous to reason directly on fibered categories without using explicit cleavages, which in particular dispenses with having to appeal, for the simple notion of $\mathcal{E}$ -functor or of cartesian $\mathcal{E}$ -functor, to a cumbersome interpretation as above. It is to avoid unbearable heaviness, and to obtain more intrinsic statements, that we have had to renounce starting as in [2] from the notion of cloven category (called “fibered category” in loc.cit.), which recedes to second rank in favor of that of fibered category. It is moreover probable that, contrary to the usage still predominant now, tied to old habits of thought, it will eventually prove more convenient in universal problems not to put the emphasis on *one* solution supposed chosen once and for all, but to put all solutions on an equal footing.

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